Tuning-Free Incremental Learning via Multilevel Prototypical Parts for Few-Shot Land Cover Segmentation

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***Abstract*—Dynamic Earth observation applications often require land-cover segmentation models to recognize newly emerging classes from only a few pixel-level annotations while retaining previously learned knowledge. However, gradient-based incremental fine-tuning (FT) is prone to severe overfitting and catastrophic forgetting under such data-scarce conditions, and conventional prototype-based methods usually rely on a single global prototype that is insufficient for complex remote sensing scenes. To address these challenges, we propose a tuning-free multilevel prototypical parts network (MLPPN) for few-shot incremental land-cover segmentation. Inspired by recognition-by-components (RBCs) theory, MLPPN first constructs a discriminative library of multilevel prototypical parts from base classes, and then registers novel classes by gradient-free clustering and prototypical imprinting with all network parameters frozen. This design mitigates parameter drift during incremental learning and captures the heterogeneous part-level composition of land-cover objects. Experiments on four remote sensing and UAV datasets under two-shot and five-shot settings show that MLPPN achieves competitive segmentation accuracy and stable inference efficiency without incremental backpropagation.**

***Index Terms*—Few-shot incremental learning (FSIL), land cover segmentation, prototypical parts, remote sensing, tuning-free.**

I. Introduction

With the rapid advancement of high-resolution Earth observation technology, remote sensing land-cover segmentation has become an important tool for time-sensitive applications such as disaster monitoring, emergency mapping, and rapid urban response [1], [2], [3], [4], [5]. In practical dynamic Earth observation scenarios, land-surface states constantly evolve, and novel categories may appear abruptly [6]. For example, after earthquakes, floods, or large-scale urban events, newly formed disaster ruins, temporary rescue shelters, blocked roads, and emergency facilities may become important target classes that are absent from the original training set. However, collecting large-scale pixel-level annotations for these emerging categories within a short response window is usually impractical because of the high cost of expert labeling and the urgency of operational deployment [7]. Therefore, deployed segmentation models should be able to incorporate novel land-cover classes from only a few annotated samples while preserving previously learned knowledge, which motivates the study of few-shot incremental land-cover segmentation [8].

Few-shot incremental land-cover segmentation is challenging because novel classes must be learned from very limited support samples while old classes remain valid in the label space. Mainstream incremental methods often rely on gradient-based fine-tuning (FT), which can become unreliable in complex remote sensing scenes [9]. On the one hand, land-cover objects exhibit large-scale variations, high intraclass variance, and complex background interference; using only two-shot or five-shot samples for backpropagation may cause the model to overfit to specific sensor perspectives, local textures, or annotation noise. On the other hand, updating the feature extractor or classifier during novel-class learning may disturb the feature manifold established by base classes, leading to catastrophic forgetting of previous knowledge [10], [11].

Prototype-based methods provide a possible way to reduce heavy FT by representing each category with feature prototypes [12], [13]. Nevertheless, directly applying conventional prototypical learning to remote sensing segmentation still has two limitations. First, a single global prototype extracted from the deepest feature layer is insufficient for heterogeneous and cross-scale land-cover categories. For instance, an airport may contain runways, grass, tarmac, buildings, and shadows, whose local patterns cannot be fully described by one averaged vector. Such coarse aggregation may obscure internal structures and produce entangled decision boundaries between base and novel classes [14]. Second, some methods still update classifiers or projection layers when registering new classes; under extreme data scarcity, these residual gradient updates can introduce interference between new-class adaptation and prior knowledge.

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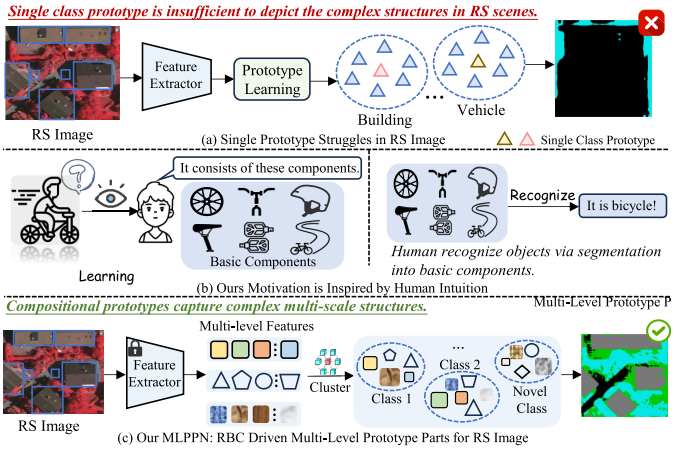


Fig. 1. Motivation of the proposed MLPPN framework. (a) Single global prototype is insufficient to depict the complex structures of remote sensing scenes. (b) Inspired by the human intuition of RBCs, where humans recognize objects by segmenting them into basic components. (c) MLPPN extracts multilevel features and clusters them into compositional prototypical parts, effectively capturing complex multiscale structures for accurate RS image segmentation.

To address these issues, as shown in Fig. 1, we propose a tuning-free multilevel prototypical parts network (MLPPN) for few-shot incremental land-cover segmentation. Inspired by recognition-by-components (RBCs) theory [15], MLPPN represents each land-cover category as a set of data-driven visual components rather than a single holistic template. Specifically, mid-level prototypical parts describe local textures, boundaries, and repeated spatial patterns, while high-level and neck-level parts encode more abstract semantic structures. During the base stage, MLPPN learns a discriminative multilevel prototypical part library through feature clustering [16], adaptive attention, and a prototypical separation loss. During the incremental stage, the backbone and attention parameters are frozen, and novel classes are registered only by forward feature extraction, gradient-free clustering, and prototype imprinting. This tuning-free mechanism reduces parameter drift, while the multilevel part representation better matches the heterogeneous composition of remote sensing scenes.

The main contributions of this article are summarized as follows.

1) A tuning-free FSIL framework inspired by RBC theory is proposed. By isolating base representation learning from novel-class registration, the framework mitigates gradient-induced catastrophic forgetting and few-shot overfitting under data-scarce conditions.
2) A multilevel prototypical parts mechanism is developed to describe heterogeneous land-cover categories with cross-scale part-level prototypes, together with a prototypical separation loss for improving feature discrimination.
3) An adaptive attention and unbiased inference strategy is designed to fuse multilevel matching responses and reduce parameter bias between base and novel classes during prototype-based prediction.
4) Extensive qualitative and quantitative evaluations are conducted on four remote sensing and UAV datasets, demonstrating competitive performance under challenging two-shot and five-shot incremental settings without incremental backpropagation.

II. Related Work

A. Semantic Segmentation of Remote Sensing Images

In recent years, deep learning has made significant progress in the field of remote sensing image processing, and methods based on convolutional neural networks and vision transformers have been widely applied to the pixel-level extraction of land cover elements [1], [2]. Early research primarily focused on improving segmentation accuracy through deep feature fusion or context modeling. However, these methods typically follow a closed-set training paradigm, assuming that the training and testing sets share the same category space. In practical Earth observation tasks, due to the dynamics of the geographical environment, new categories of land cover constantly emerge. Such static models face immense performance challenges when confronting open-world scenarios [6]. Recent advances in transformer-based architectures have shown promise for capturing long-range dependencies in images, which is particularly useful for modeling complex spatial patterns in remote sensing data [17]. Fully convolutional networks (FCNs) were among the first deep learning architectures to enable end-to-end pixel-wise prediction, laying the foundation for modern semantic segmentation in remote sensing [18]. Finally, U-Net and its variants introduced encoder–decoder structures with skip connections, significantly improving the recovery of fine-grained spatial details [19]. Self-supervised and contrastive learning approaches have further enhanced feature representation quality without requiring large labeled datasets, which is critical given the limited availability of annotated remote sensing images [20]. Residual networks and deep backbone architectures have been widely adopted to extract hierarchical features, improving segmentation accuracy in heterogeneous landscapes [21]. Meanwhile, atrous convolutions and encoder–decoder schemes allow models to capture multiscale context effectively, which is crucial for distinguishing similar land cover types [22]. Pyramid-based context aggregation strategies, such as those in PSPNet, also contribute to robust scene parsing by combining global and local information [23]. In remote sensing applications, large-scale datasets have been successfully classified using CNNs, demonstrating that deep learning can scale to high-resolution imagery and extensive geographic areas [24]. Comprehensive surveys and tutorials on deep learning for remote sensing provide practical guidelines for implementing these methods and highlight the importance of model selection, training strategies, and evaluation metrics [25]. Collectively, these developments illustrate a shift from closed-set, hand-crafted pipelines toward flexible, scalable, and context-aware deep learning models that are better equipped to handle the evolving challenges of semantic segmentation in remote sensing.

B. Class-Incremental Learning

Class-incremental learning (CIL) aims to endow models with the ability to learn new categories while retaining old knowledge. The core challenge in this field is catastrophic forgetting, where the model overwrites previously encoded

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knowledge when sequentially learning and updating parameters to adapt to new tasks [10], [26]. To alleviate this issue, existing studies mainly employ knowledge distillation techniques to constrain feature consistency between old and new models, or utilize experience replay by retaining a small number of old class samples to assist training in the incremental stage [9], [27]. However, in remote sensing land cover segmentation tasks, the spatial correlation between categories is extremely strong, and the cost of fine-grained annotation is high, making it difficult for traditional incremental FT paradigms to achieve an effective balance between plasticity and stability [28], [29].

In addition to these traditional strategies, several classical and recent methods have been proposed to further alleviate catastrophic forgetting. Elastic weight consolidation (EWC) constrains updates to parameters deemed important for previous tasks, effectively preserving old knowledge while learning new categories [30]. Knowledge distillation has also been applied in incremental learning to transfer information from previous models to maintain performance on old classes [31]. Gradient episodic memory (GEM) and tiny episodic memory (TEM) retain a small set of historical samples to guide training on new tasks and reduce forgetting [32], [33]. Meta-learning approaches, such as model-agnostic meta-learning (MAML), allow models to rapidly adapt to new categories with minimal updates, indirectly mitigating catastrophic forgetting [34].

Other strategies, including average GEM (A-GEM) and synaptic intelligence (SI), enhance the stability of incremental learning by regulating gradient directions or tracking parameter importance across tasks [35], [36]. Integrated frameworks like iCaRL combine nearest-neighbor classifiers with knowledge distillation to incrementally update both representations and classifiers [37]. More recent works address the imbalance between old and new classes during incremental updates by rebalancing loss functions or adjusting sample distributions, preventing new classes from overwhelming old classes [38]. These methods provide valuable insights for remote sensing land cover segmentation, helping maintain feature consistency, selectively retain past samples, regularize gradient updates, and rebalance class distributions while coping with strong spatial correlations and high annotation costs.

C. FSIL and Prototypical Networks

Few-shot incremental learning (FSIL) combines the dual challenges of catastrophic forgetting in continual learning and data scarcity in few-shot learning. It requires the model to expand its knowledge base to incorporate new classes using only a minimal number of annotated samples while preserving performance on base classes [8], [9]. A cornerstone of this field is the prototypical network [12], which performs classification by computing Euclidean distances to class centroids in a learned metric space. This paradigm has been extended to incremental scenarios because of its ability to define new class boundaries without extensive parameter updates [13], [39].

However, existing prototype-based segmentation schemes, especially in complex domains like remote sensing, frequently encounter representation bottlenecks. Specifically, they rely heavily on a single global prototype to represent an entire semantic category. As noted by Cermelli et al. [14], such holistic representations ignore the inherent intraclass variance and the complex composite structures within land covers (e.g., "urban" areas containing both buildings and vegetation). This leads to a severe lack of discriminative power at land cover boundaries, where the semantic transition between new and old categories becomes blurred [40].

Furthermore, a significant pitfall in many state-of-the-art methods is the continued reliance on local gradient optimization (FT) during the incremental stage. While effective in large-data regimes, gradient descent triggers severe spatial overfitting when samples are extremely scarce, causing the model's feature extractor to collapse toward the few available samples [41].

To address the aforementioned limitations, the proposed method deviates from the traditional FT paradigm. Inspired by research on visual mid-level patterns and part-level representations [14], [16], this article constructs multilevel prototypical parts and adopts a tuning-free imprinting strategy during the incremental stage, effectively solving the dilemmas of catastrophic forgetting and spatial overfitting for complex remote sensing land cover in few-shot scenarios.

III. Proposed Method

A. Overview

In FSIL, the fundamental challenge lies in enabling the model to continuously acquire novel categories using extremely scarce annotated samples while preventing catastrophic forgetting of previously learned base classes. Traditional FT-based incremental learning methods are susceptible to overfitting under data-scarce scenarios and may disrupt the old knowledge feature space. To address these challenges, this article proposes a tuning-free FSIL framework based on multilevel prototypical parts, termed MLPPN. The core motivation of this framework is that remote sensing targets often exhibit high intraclass variance and complex spatial structures; thus, directly utilizing a single global prototype to represent a complex land-cover category is typically insufficient. Therefore, this article decouples categories into multiple localized "parts" and leverages the respective advantages of mid-level features (rich in edge structures and local commonalities) and high-level features (rich in high-order abstract semantics) from deep neural networks to construct a composite representation space.

As shown in Fig. 2, the framework follows three consecutive stages: base training, tuning-free incremental learning, and collaborative inference. First, in the data-abundant base-class training stage, the model utilizes clustering algorithms to generate multilevel prototypical parts and employs the proposed joint intraclass and interclass separation loss to optimize the feature space, while simultaneously learning intralayer and interlayer attention weights. Subsequently, during the incremental stage, where only a few support samples of novel classes are available, the model adopts a tuning-free strategy. The network stops backpropagation and extracts multilevel features of the novel classes through forward inference, completing the clustering and registration of novel-class prototypical parts in the feature space. This nonparametric update

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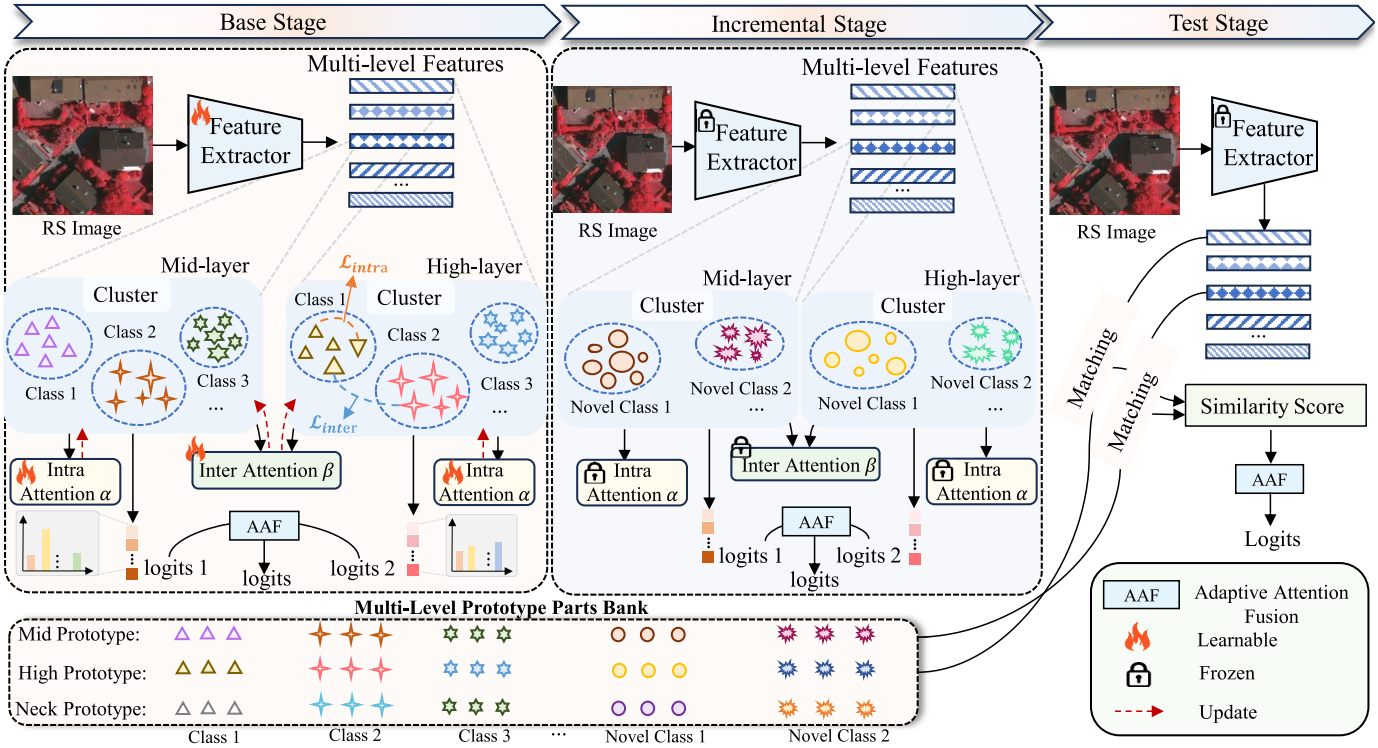


Fig. 2. Overview of the proposed MLPPN framework. In the base stage, learnable multilevel features are clustered into prototypical parts and optimized alongside intra/inter attention weights. During the incremental stage, the network and attention weights are strictly frozen; novel class prototypes are generated via forward-pass clustering and directly appended to the prototype bank without FT. In the test stage, query features are matched against the comprehensive prototype bank, and final segmentation logits are generated via adaptive attention fusion (AAF).

mechanism reduces parameter drift and alleviates catastrophic forgetting. Finally, in the deployment inference stage, based on dense distance matching and guided by the learned multilevel attention weights, the model outputs semantic segmentation masks through cross-layer collaborative decision-making.

To avoid ambiguity in the following description, we use "multilevel features" to denote the feature representations extracted from different network depths. When a specific level is discussed, we use "mid-level feature," "high-level feature," and "neck feature" consistently. For the clustered representative vectors, we uniformly use the term "prototypical part." Thus, each class is represented by a set of multilevel prototypical parts rather than by a single global prototype.

B. Base Stage: Prototype Generation and Model Training

During the base class training stage, the primary optimization objective is to sculpt a highly discriminative feature extraction network, enabling it to output multiscale feature maps with high cohesion and strong discriminability, while simultaneously establishing an attention allocation mechanism capable of measuring the importance of different parts and hierarchical levels.

Given an input image $\mathbf{X} \in \mathbb{R}^{3\times H\times W}$, the algorithm first utilizes a backbone network $\phi(\cdot)$ and a feature bottleneck layer $\psi(\cdot)$ to extract mid-level semantic features $\mathbf{F}_{mid}$ and high-level semantic features $\mathbf{F}_{high}$, respectively,

$$\mathbf{F}_{mid} = \phi(\mathbf{X}) \in \mathbb{R}^{C_{mid}\times H_{mid}\times W_{mid}} \tag{1}$$

$$\mathbf{F}_{high} = \psi(\phi(\mathbf{X})) \in \mathbb{R}^{C_{high}\times H_{high}\times W_{high}} \tag{2}$$

where C_{mid} and C_{high} represent the channel dimensions of the features at different levels. Considering that the spatial resolution of high-level features is significantly reduced after multiple pooling operations, the algorithm employs nearest-neighbor interpolation to downsample the ground truth (GT) label mask $\mathbf{Y}$ to the spatial dimensions of the corresponding feature maps, denoted as $\hat{\mathbf{Y}}_{mid}$ and $\hat{\mathbf{Y}}_{high}$, respectively, to achieve exact spatial alignment between features and labels. For any category c belonging to the base class set C_{base}, the model constructs a boolean mask $\mathbf{M}_c = \mathbb{I}(\hat{\mathbf{Y}} = c)$ via an indicator function, thereby extracting all activated pixel sets of this category on the feature map. Taking the mid-level features as an example, the pixel feature set for this category is strictly defined as

$$\mathcal{F}^c_{mid} = \{\mathbf{F}_{mid}(x,y) \mid \mathbf{M}_c(x,y) = 1\} \in \mathbb{R}^{N_c\times C_{mid}} \tag{3}$$

where N_c is the total number of pixels contained in category c within the current batch. Given the complexity of remote sensing targets, a single mean prototype struggles to capture the multimodal distribution within a class. Therefore, this article introduces the K-means clustering algorithm to perform unsupervised part-level clustering on the high-dimensional feature set $\mathcal{F}^c_{mid}$ and its corresponding high-level feature set. Ultimately, the multiple mid-level and high-level prototypical parts generated for category c can be formally expressed as

$$\mathbf{P}^c_{mid} = \text{Clustering}\left(\mathcal{F}^c_{mid}\right) = \left\{\mathbf{p}^{c,k}_{mid}\right\}_{k=1}^{K^c_{mid}} \in \mathbb{R}^{K^c_{mid}\times C_{mid}} \tag{4}$$

$$\mathbf{P}^c_{high} = \text{Clustering}\left(\mathcal{F}^c_{high}\right) = \left\{\mathbf{p}^{c,k}_{high}\right\}_{k=1}^{K^c_{high}} \in \mathbb{R}^{K^c_{high}\times C_{high}}. \tag{5}$$

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After generating prototypical parts with local representation capabilities, the network needs to evaluate the similarity response between each spatial position in the input image feature matrix and the prototype library. This article selects the normalized Euclidean distance $\mathcal{D}(\cdot,\cdot)$ as the underlying metric. Specifically, the negative distance response between the pixel feature $\mathbf{F}_{mid}(x,y)$ and the kth prototypical part $\mathbf{p}_{mid}^{c,k}$ of category c is defined as

$$\mathbf{S}_{mid}^{c,k}(x,y) = -\tau \cdot \mathcal{D}\left(\mathbf{F}_{mid}(x,y), \mathbf{p}_{mid}^{c,k}\right) \tag{6}$$

where τ is a fixed temperature scaling factor (empirically set to 10 in this article), whose core function is to smooth the distance distribution and amplify the gradient differences of distance values within the $[0,1]$ interval, thereby accelerating network convergence. Considering that different local parts within the same category contribute differently to the final classification decision (for instance, the core "roof" part of a building should naturally possess a higher decision weight than the marginal "shadow" part), this article innovatively introduces a learnable intralayer part attention weight vector $\mathbf{w}_{proto-mid}^{c} \in \mathbb{R}^{K_{mid}^{c}}$. By performing element-wise multiplication between this weight vector and the distance response, the model adaptively selects the most representative core part response for that category at the given pixel location, serving as the mid-level prediction logit

$$\mathcal{L}ogits_{mid}^{c}(x,y) = \max_{k\in\{1,\ldots,K_{mid}^{c}\}} \left(\mathbf{w}_{proto-mid}^{c,k} \cdot \mathbf{S}_{mid}^{c,k}(x,y)\right). \tag{7}$$

Similarly, semantic features of different depths play highly complementary roles in identifying complex land covers. To this end, this article defines a bilinear interpolation operator $\mathcal{U}(\cdot)$ to restore the prediction maps of each level to the original image resolution $H \times W$, and introduces interlayer attention scalar weights $w_{layer-l}$ (where $l \in \{mid, high\}$). The final pixel-wise multilevel joint prediction result is nonlinearly fused through a cross-layer max-pooling operation

$$\mathcal{L}ogits^{c}(x,y) = \max_{l\in\{mid,high\}} \left(w_{layer-l} \cdot \mathcal{U}\left(\mathcal{L}ogits_{l}^{c}\right)(x,y)\right). \tag{8}$$

To guide the network in generating prototypical parts that are both highly discriminative and possess diverse representation capabilities, this article designs a joint optimization objective consisting of three components. The first is the standard cross-entropy loss $\mathcal{L}_{cls}$, used to guide the global alignment between model predictions and GT labels. Second, to prevent different prototypical parts within the same category from undergoing feature degradation and collapsing into the same point in the feature space during optimization, this article designs an intraclass prototypical part separation loss $\mathcal{L}_{intra}$. This loss aims to maximize the distance between any two different prototypical parts within the same category

$$\mathcal{L}_{intra} = \sum_{l\in\{mid,high\}} \frac{1}{|\mathcal{C}_{base}|} \sum_{c\in\mathcal{C}_{base}} \left(\frac{-1}{K_l^c\left(K_l^c-1\right)} \sum_{i\neq j} \mathcal{D}\left(\mathbf{p}_l^{c,i}, \mathbf{p}_l^{c,j}\right)\right). \tag{9}$$

Finally, to further broaden the discriminative boundaries between different categories in the feature space, this article calculates the geometric center of the prototypical parts for each category c at the corresponding level, denoted as $\bar{\mathbf{p}}_l^c = 1/K_l^c \sum_{k=1}^{K_l^c} \mathbf{p}_l^{c,k}$, and proposes an interclass prototype separation loss $\mathcal{L}_{inter}$ to maximize the distance between different category centers

$$\mathcal{L}_{inter} = \sum_{l\in\{mid,high\}} \frac{-1}{|\mathcal{C}_{base}|\left(|\mathcal{C}_{base}|-1\right)} \sum_{c_i\neq c_j} \mathcal{D}\left(\bar{\mathbf{p}}_l^{c_i}, \bar{\mathbf{p}}_l^{c_j}\right). \tag{10}$$

In summary, the overall optimization objective for the base training stage is the weighted sum of the three aforementioned loss functions, i.e., $\mathcal{L} = \mathcal{L}_{cls} + \lambda_1\mathcal{L}_{intra} + \lambda_2\mathcal{L}_{inter}$.

Algorithm 1 Tuning-Free Incremental Learning of MLPPN

Input: base training set $\mathcal{D}_{base}$, incremental support sets $\{\mathcal{S}_t\}_{t=1}^{T}$, query image $\mathbf{X}_q$
Base stage: train $\phi(\cdot)$, $\psi(\cdot)$, intra-layer part attention weights, and inter-layer attention weights on $\mathcal{D}_{base}$
Extract multi-level base-class features and cluster pixel features into base-class prototypical parts
Optimize the model with the segmentation loss and prototype-separation losses
Incremental stage: freeze $\phi(\cdot)$, $\psi(\cdot)$, and all learned attention weights
for each incremental support set $\mathcal{S}_t$ **do**
    Extract mid-level and high-level support features by forward inference only
    Collect labeled pixels for each novel class and cluster them into multi-level prototypical parts
    Initialize novel-class part attention weights from base-class attention statistics
    Append the novel-class prototypical parts to the global prototype bank
end for
Inference stage: extract query features with the frozen network
Match query features against all base/novel prototypical parts and fuse multi-level logits
Predict $\mathbf{M}_{pred}$ by pixel-wise argmax
Output: segmentation prediction $\mathbf{M}_{pred}$

C. Incremental Stage: Tuning-Free Novel Class Learning

Algorithm 1 summarizes the tuning-free incremental learning procedure of MLPPN. It explicitly separates the trainable base stage, the frozen incremental registration stage, and the nonparametric inference stage.

Following the convergence of the base class model training, the MLPPN framework transitions into the FSIL sessions. In the tth incremental session, the model encounters a standard N-way K-shot task setting, meaning the model only has access to a small-scale support set containing N entirely new land cover categories (forming the novel class set $\mathcal{C}_{novel}^{t}$). This support set can be strictly defined as $\mathcal{S}_t = \{(\mathbf{X}_{t,i}, \mathbf{Y}_{t,i})\}_{i=1}^{N\times K}$, where each image $\mathbf{X}_{t,i}$ only provides extremely scarce pixel-level annotations. Under such severe data scarcity, employing

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traditional gradient descent for network parameter FT would cause the drastic gradient updates of novel classes to instantly wash away the smooth feature manifolds established during the base stage, resulting in irreversible catastrophic forgetting.

To address this dilemma, MLPPN adopts a tuning-free paradigm. At the onset of the incremental stage, all learnable parameters of the feature extraction network, including the backbone $\phi(\cdot)$ and the bottleneck layer $\psi(\cdot)$, are frozen, and the intralayer and interlayer attention weights learned during the base stage are kept fixed. The incremental learning process is therefore transformed into feature-space projection and prototype registration. Specifically, the algorithm inputs the images from the novel-class support set $\mathcal{S}_t$ one by one into the frozen network, extracting the corresponding novel-class mid-level features $\mathbf{F}_{mid}^{t,i}$ and high-level features $\mathbf{F}_{high}^{t,i}$ via forward propagation. Subsequently, using the GT label masks $\mathbf{Y}_{t,i}$ provided by the support set, the algorithm extracts the pixel features belonging to each novel category $n \in \mathcal{C}_{novel}^{t}$ on the multilevel feature maps. Taking mid-level features as an example, this novel-class feature set is aggregated as

$$\mathcal{F}_{mid}^{n} = \bigcup_{i=1}^{N\times K} \left\{ \mathbf{F}_{mid}^{t,i}(x,y) \;\middle|\; \mathbf{Y}_{t,i}(x,y) = n \right\}. \tag{11}$$

After completing the feature aggregation, the algorithm reuses the unsupervised K-means clustering mechanism from the base stage to cluster the high-dimensional feature set $\mathcal{F}_{mid}^{n}$ and its corresponding high-level feature set, thereby generating the multilevel prototypical part sets $\mathbf{P}_{mid}^{n}$ and $\mathbf{P}_{high}^{n}$ for the novel category n. This equips the representation of novel categories with the same capacity to handle high intraclass variance. Subsequently, these newly generated prototypical parts are concatenated into the global prototype bank, expanding the set of recognizable categories without updating the network parameters

$$\mathbf{P}_{mid}^{all} = \mathbf{P}_{mid}^{base} \cup \mathbf{P}_{mid}^{novel_1} \cup \cdots \cup \mathbf{P}_{mid}^{novel_t}. \tag{12}$$

However, because there is no gradient backpropagation during the incremental stage, the network cannot acquire the intralayer part attention weights $\mathbf{w}_{proto-mid}^{n}$ for novel categories through end-to-end learning as it did in the base stage. Considering that the feature manifolds extracted by the same network inherently share distributional consistency, this article proposes a prior-knowledge-based weight initialization strategy, utilizing the statistical mean of the base class attention weights to unbiasedly initialize the novel class weights. This process can be mathematically expressed as

$$\mathbf{w}_{proto-mid}^{n} = \frac{1}{|\mathcal{C}_{base}|} \sum_{c\in\mathcal{C}_{base}} \mathbf{w}_{proto-mid}^{c} \in \mathbb{R}^{K_{mid}^{n}}. \tag{13}$$

Through the matrix operations and forward inference described above, MLPPN can efficiently complete the registration of novel categories. This nonparametric incremental mechanism enables efficient novel-class registration without backpropagation and helps maintain the base-class feature space during incremental learning.

D. Inference Stage: Multilevel Collaborative Decision

Upon the completion of base training and several incremental sessions, the model proceeds to the testing and inference stage for practical deployment. At this point, given any unknown remote sensing query image $\mathbf{X}_q$, the algorithm no longer distinguishes between base and novel classes; instead, it relies on a unified multilevel collaborative decision pipeline to generate the final dense semantic prediction mask $\mathbf{M}_{pred}$.

First, the query image $\mathbf{X}_q$ is fed into the dual-branch feature extraction network, which remains in a parameter-frozen state, to stably and deterministically output its corresponding multiscale feature tensors $\mathbf{F}_{mid}^{q}$ and $\mathbf{F}_{high}^{q}$. For any spatial coordinate (x, y) on the feature map, the model computes the normalized Euclidean distance between this pixel feature and every prototypical part in the current global prototype libraries $\mathbf{P}_{mid}^{all}$ and $\mathbf{P}_{high}^{all}$ (which now contain all prototypical parts of both base and novel classes). For any category $c \in \mathcal{C}_{all}$ (where $\mathcal{C}_{all} = \mathcal{C}_{base} \cup \mathcal{C}_{novel}$) within the global category set, its negative distance response at the kth part in the mid-level feature space is rigorously calculated as

$$d_{mid}^{c,k}(x,y) = \mathcal{D}\left(\mathbf{F}_{mid}^{q}(x,y), \mathbf{p}_{mid}^{c,k}\right). \tag{14}$$

Subsequently, under the modulation of the solidified intralayer part attention weights (comprising the learned weights of base classes and the inherited mean weights of novel classes), the network performs weighted filtering on these massive distance metrics. For each specific category c, the algorithm adaptively filters out the core part with the strongest matching response internally, utilizing it as the relative confidence score supporting that category at this level

$$\mathcal{L}ogits_{mid}^{c}(x,y) = \max_{k\in\{1,\ldots,K_{mid}^{c}\}} \left(-\tau \cdot \mathbf{w}_{proto-mid}^{c,k} \cdot d_{mid}^{c,k}(x,y)\right). \tag{15}$$

Due to spatial scale discrepancies between mid-level features (which preserve high-frequency boundaries) and high-level features (which embed low-frequency semantics), their respective score maps $\mathcal{L}ogits_{mid}^{c}$ and $\mathcal{L}ogits_{high}^{c}$ cannot be directly compared. Therefore, the algorithm utilizes the bilinear interpolation operator $\mathcal{U}(\cdot)$ to precisely upsample both sets of score maps to the original image resolution $H \times W$. Building upon this, the network incorporates the interlayer attention scalar weights $w_{layer-mid}$ and $w_{layer-high}$ for importance scaling, accomplishing a secondary voting fusion of multiscale information via a pixel-wise cross-layer max-pooling operation. This process constructs the final full-image comprehensive response tensor $\mathcal{L}ogits^{c}$. Let $F_{l}^{c}(x,y) = w_l \cdot \mathcal{U}(\mathcal{L}_{l}^{c})(x,y)$ be the weighted upsampled logits at level $l \in \{\text{mid}, \text{high}\}$, the fusion objective is

$$\mathcal{L}ogits^{c}(x,y) = \max\left(F_{\text{mid}}^{c}(x,y),\ F_{\text{high}}^{c}(x,y)\right). \tag{16}$$

Finally, on the fused joint response tensor, the algorithm executes a discrete Argmax operation along the category channel dimension. For every pixel point (x, y) on the image, the model strictly selects the category identifier with the highest joint confidence as the sole classification result, thereby generating an accurate 2-D semantic segmentation mask

$$\mathbf{M}_{pred}(x,y) = \arg\max_{c\in\mathcal{C}_{all}} \mathcal{L}ogits^{c}(x,y). \tag{17}$$

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Relying on this nonparametric inference pipeline based on dense distance metrics and multilevel part comparisons, MLPPN reduces the parameter bias problem commonly observed in traditional classifier heads during the incremental process. Even when faced with real-world remote sensing environments characterized by scale variations of land covers and complex background interference under few-shot scenarios, this mechanism supports balanced collaborative discrimination for both old and new categories.

IV. Experiments

A. Datasets

To comprehensively evaluate the effectiveness of the proposed method in handling complex environments and varied spatial resolutions, extensive experiments are conducted on four mainstream semantic segmentation datasets.

The aeroscapes dataset [42] is a specialized computer vision benchmark for semantic segmentation, featuring images captured entirely by uncrewed aerial vehicles (UAVs) flying at low altitudes ranging from 5 to 50 m. It contains 3269 finely annotated images with a high resolution of 1280×720 pixels. The dataset encompasses typical outdoor and suburban scenes, defining 11 core foreground categories alongside a background class. These categories include person, bicycle, car, drone, boat, animal, obstacle, structure, vegetation, road, and sky.

The dense labeling remote sensing dataset (DLRSD) [43], [44] is a high-resolution densely annotated remote sensing dataset released by research institutions including Wuhan University. It serves as a significant pixel-level extension of the renowned UC Merced Land Use dataset. The dataset comprises 2100 images with a spatial resolution of approximately 0.3 m and a unified size of 256×256 pixels, distributed evenly across 21 scene categories. At the pixel level, the dataset provides dense annotations for 17 diverse land cover classes, encompassing airplane, bare soil, building, car, chaparral, court, dock, field, grass, mobile home, pavement, sand, sea, ship, tank, tree, and water.

The ISPRS Potsdam and Vaihingen Semantic Labeling Datasets are renowned benchmarks for urban semantic segmentation in the remote sensing domain [45]. The Potsdam dataset consists of 38 large-scale true orthophotos covering typical European urban landscapes, acquired by airborne sensors with a ground sampling distance of 5 cm and an original resolution of 6000×6000 pixels. Similarly, the Vaihingen dataset provides high-resolution aerial imagery with a ground sampling distance of 9 cm. Both datasets define six representative urban land cover categories including impervious surfaces, buildings, low vegetation, trees, cars, and background. During the experimental phase, the original large-scale images from both datasets are cropped into smaller patches of 300×300 pixels for training and testing purposes.

B. Experimental Settings

The implementation of the proposed method strictly follows the configurations of the baseline comparison method PIFS to ensure a fair evaluation. The network architecture adopts ResNet101 as the backbone for robust feature extraction and utilizes DeepLabV3 + as the bottleneck module. The backbone is initialized with weights pretrained on the ImageNet dataset. The entire network is optimized using the Adam optimizer with an initial learning rate of 0.001. A polynomial learning rate decay strategy (PolyLR) is applied with a step size of 4 and a gamma value of 0.9 to dynamically adjust the learning rate during training.

Regarding the training configurations for the base phase sessions across different datasets, the mini-batch sizes are empirically set to 2 for aeroscapes, 36 for DLRSD, 16 for Potsdam, and 10 for Vaihingen. The base models are trained for 200 epochs on aeroscapes and 150 epochs on DLRSD, Vaihingen, and Potsdam. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. Unless otherwise specified, the main results are obtained with random seed 42. Ultimately, the model checkpoint achieving the highest performance on the validation set is selected and utilized for the final testing phase.

It is worth noting that the five-shot setting does not always lead to a clear improvement over the two-shot setting in every dataset and session. This phenomenon is reasonable in few-shot incremental segmentation because performance depends not only on the number of support samples but also on their spatial coverage, category representativeness, and annotation quality. If the additional support images mainly contain redundant local appearances or include visually confusing background regions, the clustered prototypical parts may shift toward biased support distributions. Moreover, small objects and spectrally similar land-cover categories can make the incremental prototypes less stable. Therefore, the comparison between two-shot and five-shot should be interpreted together with the class difficulty and support-sample distribution, rather than as a strictly monotonic trend.

C. Comparison With State-of-the-Arts

To comprehensively evaluate the effectiveness of our proposed method, we compare it against a wide range of representative baselines and state-of-the-art approaches. We divide these compared methods into three main categories. First, we consider general baseline and classical low-shot/incremental learning techniques, including standard FT [46], learning without forgetting (LWF) [11], weight imprinting (WI) [47], dynamic WI (DWI) [48], and rethinking (RT) [41]. Second, we compare with several prominent methods designed for class-incremental semantic segmentation (CISS), namely, ILT [49], MIB [40], PLOP [50], MBS [51], and NEST [52]. Finally, we benchmark against algorithms specifically tailored for few-shot and incremental few-shot semantic segmentation, including adaptive masked proxies (AMPs) [53], PIFS [54], GAPS [55], and MCLOS [56].

1) Results on the Aeroscapes Dataset: To validate the continual learning capability of the proposed MLPPN under extreme few-shot conditions, we conduct rigorous incremental semantic segmentation experiments on the aeroscapes aerial remote sensing dataset, which features highly complex backgrounds. As summarized in Tables I and II, under the 8–1–1–1–1 incremental setting for both two-shot and five-shot scenarios, the quantitative results significantly demonstrate the

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TABLE I

MIOU STATISTICS FOR EACH STAGE OF A **TWO-SHOT** SESSION ON THE AEROSCAPES DATASET (8–1–1–1–1)

Algorithm	$mIoU_{base}$				$mIoU_{novel}$				$mIoU_{all}$			
	S1	S2	S3	S4	S1	S2	S3	S4	S1	S2	S3	S4
AMP	40.02	32.91	28.11	26.36	9.67	11.87	11.06	12.46	24.84	22.39	19.58	19.41
DWI	**47.73**	40.87	32.83	32.68	0.00	0.00	0.00	0.00	23.87	20.43	16.42	16.34
FT	12.33	9.10	5.27	2.25	0.00	14.60	6.81	9.78	6.17	11.85	6.04	6.02
ILT	11.73	5.46	5.56	1.43	0.00	6.03	6.19	10.81	5.87	5.75	5.88	6.12
LWF	11.81	6.47	4.73	1.64	0.00	4.80	1.69	12.19	5.90	5.63	3.21	6.91
MCLOS	9.19	3.87	3.07	2.94	0.94	0.00	0.00	0.00	5.07	1.94	1.54	1.47
MIB	11.71	5.58	9.15	3.39	0.00	27.74	4.27	1.65	5.86	16.66	6.71	2.52
PIFS	14.28	3.87	3.07	2.21	2.93	0.02	0.02	1.65	8.60	1.94	1.54	1.93
RT	11.96	6.53	4.26	3.77	0.00	24.70	16.50	12.38	5.98	15.62	10.38	8.08
WI	46.29	40.23	33.07	**32.85**	17.24	7.19	6.26	9.16	31.77	23.71	19.66	21.01
NEST	43.70	37.21	32.71	32.47	0.00	0.00	0.00	0.00	21.85	18.60	16.36	16.24
PLOP	47.63	27.20	18.24	19.40	0.00	3.44	1.62	1.15	23.82	15.32	9.93	10.28
MBS	36.57	25.90	22.70	19.36	0.00	36.22	**36.95**	21.97	18.28	31.06	**29.82**	20.66
GAPS	44.60	**41.08**	**36.37**	31.86	0.12	0.16	0.02	0.41	22.36	20.62	18.19	16.13
MLPPN	43.84	40.67	30.42	30.32	**38.41**	**37.86**	28.32	**24.84**	**41.12**	**39.26**	29.37	**27.58**

TABLE II

MIOU STATISTICS ACROSS SESSION STAGES FOR **FIVE-SHOT** ON THE AEROSCAPES DATASET (8–1–1–1–1)

Algorithm	$mIoU_{base}$				$mIoU_{novel}$				$mIoU_{all}$			
	S1	S2	S3	S4	S1	S2	S3	S4	S1	S2	S3	S4
AMP	39.87	32.80	27.95	26.18	9.00	11.91	11.12	12.42	24.43	22.36	19.53	19.30
DWI	**47.73**	40.87	32.84	**32.68**	0.00	0.00	0.00	0.00	23.87	20.43	16.42	16.34
FT	12.45	8.14	8.18	3.29	0.00	6.47	7.27	10.88	6.23	7.30	7.73	7.08
ILT	12.59	11.27	6.61	4.24	0.00	5.08	7.20	13.98	6.30	8.17	6.90	9.11
LWF	12.84	5.71	4.60	2.91	0.00	3.51	2.83	5.88	6.42	4.61	3.71	4.39
MCLOS	9.19	3.87	3.07	2.94	0.94	0.00	0.00	0.00	5.07	1.94	1.54	1.47
MIB	13.41	0.68	3.61	1.22	0.01	29.11	14.80	14.55	6.71	14.89	9.20	7.88
PIFS	10.95	3.87	3.02	0.01	0.23	0.00	0.52	11.00	5.59	1.94	1.77	5.50
RT	13.38	5.09	6.33	4.73	0.00	28.36	5.53	3.94	6.69	16.72	5.93	4.33
WI	46.31	40.24	33.06	28.84	18.28	7.30	6.30	9.19	32.29	23.77	19.68	19.02
NEST	43.70	37.21	32.71	32.46	0.01	0.00	0.00	0.00	21.86	18.60	16.35	16.23
PLOP	47.63	27.20	18.24	19.40	0.06	3.44	1.62	1.15	23.85	15.32	9.93	10.28
MBS	36.72	26.13	22.88	19.48	0.00	**36.67**	**37.27**	22.25	18.36	31.40	**30.07**	20.86
GAPS	44.75	**41.28**	**36.49**	31.94	0.17	0.26	0.12	0.61	22.46	20.77	18.30	16.27
MLPPN	44.60	41.18	30.77	30.64	**40.03**	36.59	27.55	**24.09**	**42.32**	**38.88**	29.16	**27.36**

superiority of MLPPN. Regarding the overall performance ($mIoU_{all}$), as the incremental stages progress, the vast majority of baseline methods suffer from severe performance collapse, whereas MLPPN maintains an absolute leading position across all incremental sessions. For instance, in the final stage (Session 4) of the two-shot setting, the $mIoU_{all}$ of MLPPN reaches 27.58%, substantially outperforming the runner-up models such as MBS and WI. This indicates that when handling long-sequence incremental tasks for remote sensing features, our method can cope with performance degradation more gracefully, exhibiting exceptional robustness and feature retention capabilities.

Notably, in the experimental results of the baseline models, the prediction accuracy for novel classes ($mIoU_{novel}$) frequently exhibits zero or extremely low values (e.g., methods like DWI, PLOP, and NEST show large areas of 0.00%). This phenomenon profoundly reveals the core difficulty of few-shot incremental segmentation in remote sensing imagery. Remote sensing images generally possess characteristics such as huge variations in object scales, high interclass feature similarities, and extremely complex backgrounds. Under the extreme conditions of having only two or five samples, the feature extraction networks of traditional incremental models are easily overwhelmed by the vast background information, failing to effectively decouple the foreground features of novel targets. Due to the lack of sufficient supervision signals, these models tend to directly misclassify the newly introduced object categories as the old background class (i.e., triggering a severe background shift problem). Furthermore, when some baseline models forcibly utilize regularization or

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TABLE III

RECALL AND PRECISION STATISTICS ACROSS SESSION STAGES FOR **TWO-SHOT** AND **FIVE-SHOT** ON THE AEROSCAPES DATASET

Algorithm	2-shot								5-shot							
	Recall				Precision				Recall				Precision			
	S1	S2	S3	S4	S1	S2	S3	S4	S1	S2	S3	S4	S1	S2	S3	S4
AMP	31.35	31.78	31.29	38.52	64.07	49.78	42.47	41.26	31.01	31.75	31.32	38.45	69.78	49.09	42.00	40.99
DWI	29.43	29.43	29.43	29.43	49.03	59.37	41.04	36.29	29.43	29.43	29.43	29.43	49.13	59.83	41.43	36.58
FT	11.53	17.68	13.46	24.55	13.81	49.94	29.47	31.53	9.90	12.87	14.66	24.85	18.25	37.59	34.22	26.12
ILT	8.76	13.28	16.18	21.56	11.22	31.55	26.22	23.12	9.13	14.96	15.06	32.30	19.17	36.31	31.69	19.14
LWF	8.20	11.73	9.22	21.99	15.93	37.39	28.51	20.47	8.61	10.99	9.90	23.72	20.04	26.65	36.73	14.99
MCLOS	6.70	6.25	6.22	6.04	13.83	1.94	1.54	1.47	6.92	6.31	6.24	6.11	14.22	1.98	1.63	1.55
MIB	8.37	23.67	18.11	18.55	14.71	31.00	15.10	12.46	9.51	25.00	18.49	17.90	23.90	35.06	22.31	15.00
PIFS	10.29	6.26	6.25	14.29	46.46	19.75	11.29	4.48	7.24	6.25	6.37	12.51	43.94	2.89	7.81	12.84
RT	9.67	22.67	18.39	13.95	7.10	21.82	18.59	16.31	12.27	27.61	14.96	10.89	17.51	27.01	23.76	17.03
WI	36.01	30.28	29.63	32.87	**72.39**	49.72	41.44	40.20	36.59	30.34	29.64	32.79	**72.56**	49.94	41.38	40.23
NEST	53.13	41.13	31.50	30.00	25.42	25.42	25.42	25.42	53.55	41.25	31.53	30.06	25.42	25.42	25.42	25.42
PLOP	38.21	24.89	13.23	14.47	30.83	42.57	35.54	31.26	38.21	24.89	13.23	14.47	30.83	42.57	35.54	31.26
MBS	32.45	39.48	45.40	**42.15**	20.36	39.18	37.65	35.98	32.66	40.02	44.78	**42.44**	20.55	39.65	38.00	36.23
GAPS	34.89	43.20	32.04	39.73	31.63	29.26	25.19	21.95	35.01	43.45	32.19	39.91	31.73	29.41	25.29	22.00
MLPPN	**62.06**	**47.20**	**46.45**	40.59	62.38	**64.99**	**50.72**	**48.10**	**61.79**	**46.30**	**46.15**	40.16	63.50	**65.77**	**51.10**	**48.33**

TABLE IV

MIOU STATISTICS FOR EACH STAGE OF A **TWO-SHOT** SESSION ON THE DLRSD DATASET (13–1–1–1–1–1)

Method	$mIoU_{base}$					$mIoU_{novel}$					$mIoU_{all}$				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
AMP	61.27	53.69	45.14	37.58	37.28	40.70	**63.07**	35.91	24.97	23.46	50.98	**58.38**	40.52	31.27	30.37
DWI	45.55	37.95	37.57	33.89	33.73	39.57	54.66	38.37	28.85	23.99	42.56	46.30	37.97	31.37	28.86
FT	24.93	9.75	17.60	22.87	2.95	5.19	1.23	1.57	5.59	1.46	15.06	5.49	9.59	14.23	2.21
ILT	31.05	13.05	18.60	13.56	7.59	2.83	0.93	0.83	4.93	1.36	16.94	6.99	9.71	9.24	4.47
LWF	13.04	5.50	11.42	6.11	4.15	2.80	0.89	1.21	3.34	1.97	7.92	3.20	6.32	4.72	3.06
MCLOS	0.34	0.00	0.00	0.00	0.00	5.44	0.90	0.00	0.00	0.00	2.89	0.45	0.00	0.00	0.00
MIB	20.99	9.15	13.10	13.81	7.10	1.06	4.58	1.70	4.26	1.83	11.03	6.86	7.40	9.03	4.47
PIFS	0.41	0.00	1.30	0.03	0.00	3.51	0.90	1.64	3.54	0.83	1.96	0.45	1.47	1.78	0.42
RT	12.75	11.94	26.33	25.44	10.29	2.77	0.44	0.38	4.36	0.45	7.76	6.19	13.36	14.90	5.37
WI	45.21	37.61	37.19	34.92	34.77	40.62	55.18	38.43	**31.10**	24.05	42.91	46.40	37.81	33.01	29.41
NEST	54.01	46.41	46.11	43.05	42.91	40.04	54.82	**40.11**	29.25	17.30	47.02	50.62	43.11	36.15	30.10
PLOP	65.46	60.16	**59.86**	**55.60**	53.92	0.00	0.00	0.00	0.00	0.00	32.73	30.08	29.93	27.80	26.96
MBS	31.02	33.29	33.38	33.38	32.88	0.00	0.00	28.78	27.01	19.12	15.51	16.65	31.08	30.19	26.00
GAPS	57.90	54.93	50.40	47.91	45.26	5.04	0.14	3.78	21.46	2.14	31.47	27.54	27.09	34.68	23.70
MLPPN	**68.13**	**60.58**	59.63	55.06	**54.44**	**41.59**	52.96	37.83	29.72	**24.33**	**54.86**	56.77	**48.73**	**42.39**	**39.38**

knowledge distillation to preserve base class features, it leads to extreme ossification of the feature space, preventing novel classes from acquiring effective decision boundaries within the model. In contrast, MLPPN successfully breaks this zero-value trap caused by background overfitting and representation ossification. Its $mIoU_{novel}$ reaches as high as 38.41% in the first incremental stage, demonstrating outstanding feature plasticity and novel class mining capability.

In terms of maintaining base class knowledge ($mIoU_{base}$) and overall prediction reliability, MLPPN also achieves a balance in the stability-plasticity dilemma. It is observed in the experiments that although some methods (such as GAPS and DWI) retain a relatively high $mIoU$ on base classes, this is often a zero-sum game at the expense of completely losing the ability to learn novel classes. MLPPN not only performs excellently on novel classes but also maintains its base class accuracy at a high level, achieving effective compatibility within the feature space. Further combining the recall and precision metrics shown in Table III, the complex environments in remote sensing images easily lead to missed detections of objects. However, MLPPN exhibits significantly higher recall rates than the baselines across most stages, implying that it can maximally capture the complete regions of target objects. Meanwhile, its steadily increasing precision in subsequent incremental stages indicates that while actively recalling targets, the model effectively suppresses false-positive predictions brought by complex backgrounds, ensuring that the mined positive samples have high reliability.

2) Results on the DLRSD Dataset: To evaluate the performance of MLPPN in long-sequence and fine-grained land cover classification tasks, we conducted a five-stage few-shot incremental semantic segmentation experiment under the 13–1–1–1–1–1 setting on the DLRSD remote sensing dataset. The quantitative results (Tables IV and V) indicate that MLPPN possesses relatively stable continual learning capabilities when handling complex urban and land-use remote sensing scenarios. Regarding the overall performance, as the incremental stages progress from Session 1 to Session 5, most

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TABLE V

MIOU STATISTICS ACROSS SESSION STAGES FOR **FIVE-SHOT** ON THE DLRSD DATASET (13–1–1–1–1–1)

Method	$mIoU_{base}$					$mIoU_{novel}$					$mIoU_{all}$				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
AMP	59.69	52.24	46.13	38.74	38.54	42.81	**60.44**	37.36	24.62	**23.13**	51.25	56.34	41.75	31.68	30.84
DWI	45.12	37.52	36.47	33.18	33.06	37.42	53.59	37.12	27.22	21.95	41.27	45.55	36.80	30.20	27.50
FT	39.72	16.44	37.43	19.49	15.80	10.54	0.64	4.05	4.91	0.93	25.13	8.54	20.74	12.20	8.37
ILT	40.29	16.13	37.85	18.70	14.86	9.14	0.63	4.01	5.63	0.98	24.71	8.38	20.93	12.17	7.92
LWF	37.27	14.49	31.84	12.63	9.35	5.27	1.17	3.67	4.66	1.08	21.27	7.83	17.76	8.64	5.21
MCLOS	0.01	0.00	0.00	0.00	0.00	3.57	0.90	0.00	0.00	0.00	1.79	0.45	0.00	0.00	0.00
MIB	43.98	24.47	26.54	17.82	17.52	8.97	2.41	1.44	4.92	0.49	26.47	13.44	13.99	11.37	9.00
PIFS	0.58	0.00	0.00	0.86	0.50	2.97	0.90	0.32	2.80	0.99	1.77	0.45	0.16	1.83	0.75
RT	17.54	14.55	36.53	21.15	14.86	3.40	1.28	0.84	6.54	0.41	10.47	7.91	18.68	13.85	7.63
WI	44.89	37.29	36.17	33.90	33.78	38.62	54.19	37.47	26.59	21.68	41.76	45.74	36.82	30.24	27.73
NEST	54.01	46.41	44.39	35.05	34.90	40.32	54.97	37.17	26.28	15.00	47.16	50.69	40.78	30.66	24.95
PLOP	65.46	60.16	**59.86**	**55.60**	53.92	0.00	0.00	0.00	0.00	0.00	32.73	30.08	29.93	27.80	26.96
MBS	31.02	33.29	33.38	33.38	32.88	0.00	0.00	28.78	27.01	19.12	15.51	16.65	31.08	30.19	26.00
GAPS	57.90	54.93	50.40	47.91	45.26	5.04	0.14	3.78	21.46	2.14	31.47	27.54	27.09	34.68	23.70
MLPPN	**68.16**	**60.60**	58.88	54.95	**54.36**	**44.77**	55.11	**38.52**	**27.26**	22.37	**56.46**	**57.86**	**48.70**	**41.10**	**38.36**

TABLE VI

RECALL AND PRECISION STATISTICS ACROSS SESSION STAGES FOR **TWO-SHOT** AND **FIVE-SHOT** ON THE DLRSD DATASET

Method	**2-shot**										**5-shot**									
	Recall					Precision					Recall					Precision				
	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5
AMP	**80.06**	**79.05**	**71.22**	**58.19**	53.71	69.75	65.87	46.45	38.23	39.86	**83.56**	**80.36**	**73.22**	**59.59**	**55.94**	63.57	63.42	47.53	38.06	39.40
DWI	71.50	70.79	60.26	51.40	46.39	53.81	56.95	48.85	46.98	47.16	71.86	70.84	61.09	51.61	46.81	52.34	56.15	48.29	46.39	46.94
FT	48.35	27.01	27.15	31.21	14.69	29.17	24.80	22.67	22.85	15.16	62.54	21.85	39.61	27.19	20.75	36.21	27.11	28.34	32.60	24.48
ILT	43.68	26.16	29.26	18.61	12.12	33.99	23.52	24.42	24.42	22.69	61.44	21.68	40.01	27.04	20.14	35.61	27.90	28.49	39.09	25.38
LWF	38.16	26.19	26.04	15.19	14.41	32.69	24.17	19.44	19.90	24.43	62.38	30.46	35.28	25.38	18.63	34.51	27.54	28.57	32.20	23.05
MCLOS	31.95	25.00	0.00	0.00	0.00	3.00	0.45	0.00	0.00	0.00	49.97	24.99	0.00	0.00	0.00	1.97	3.57	0.00	0.00	0.00
MIB	20.47	20.53	20.71	17.46	11.85	26.25	20.06	24.15	23.11	22.15	49.75	33.07	34.09	29.17	22.32	35.26	23.65	24.57	24.81	20.64
PIFS	45.74	25.00	7.63	12.49	10.00	6.40	0.45	4.84	5.08	0.42	35.73	25.00	16.67	10.03	10.31	1.84	0.45	0.16	3.95	7.03
RT	38.62	14.32	28.92	28.35	10.83	29.38	18.77	28.62	28.02	19.10	45.69	27.69	32.73	25.10	14.93	34.88	32.98	30.88	32.40	28.23
WI	72.85	71.46	60.11	52.92	47.96	53.88	56.89	48.63	50.90	50.39	73.00	71.44	61.02	51.85	47.30	52.71	56.30	48.27	45.99	46.74
NEST	53.90	56.92	49.92	50.29	43.75	76.06	**75.19**	**64.48**	**56.08**	46.78	54.04	56.99	48.56	37.69	33.04	76.09	75.21	62.19	49.32	40.39
PLOP	37.39	34.74	34.54	31.76	30.84	38.61	38.61	38.61	38.61	38.61	37.39	34.74	34.54	31.76	30.84	38.61	38.61	38.61	38.61	38.61
MBS	27.37	24.95	43.12	46.49	40.42	19.49	20.72	39.35	37.94	32.97	27.37	24.95	43.12	46.49	40.42	19.49	20.72	39.35	37.94	32.97
GAPS	37.33	34.63	36.24	48.81	36.31	42.65	33.77	37.20	44.30	30.21	37.33	34.63	36.24	48.81	36.31	42.65	33.77	37.20	44.30	30.21
MLPPN	61.13	71.19	64.03	56.99	**55.02**	**82.85**	73.28	61.98	54.07	**55.88**	63.01	72.14	64.04	56.60	54.78	**82.47**	**76.54**	**64.76**	**57.38**	**59.41**

baseline methods exhibit significant performance degradation, whereas MLPPN maintains a leading position in the overall mean intersection over union ($mIoU_{all}$) across all stages. For instance, in the final stage (Session 5) of the two-shot and five-shot settings, the $mIoU_{all}$ of MLPPN reaches 39.38% and 38.36%, respectively, outperforming other comparative methods. This result indicates that after undergoing multiple model updates, MLPPN can better balance the learning of new knowledge and the retention of old knowledge, mitigating the trend of performance decay.

Combining the statistical data of recall and precision (Table VI), MLPPN maintains a relatively balanced performance between recall and precision during the long-sequence incremental process. In remote sensing segmentation tasks, this indicates that the model reduces missed detections of target objects while also controlling false predictions caused by similar textures. Such stability in these metrics helps the model maintain relatively accurate positive sample mining capabilities across multiple incremental iterations.

When observing the specific category-wise quantitative results of the baseline models, a large number of "0.00" values can be found in the per-category IoU statistics (Table VII) and the novel class accuracy ($mIoU_{novel}$). This phenomenon reflects the objective difficulty of FSIL in fine-grained remote sensing images. The DLRSD dataset contains many land cover categories with similar visual features. Under the condition of extremely few samples, traditional incremental learning models struggle to establish feature boundaries with sufficient discriminability. Consequently, when novel class samples are input, their feature responses are easily overshadowed by background classes or other base class features, resulting in obvious category confusion and background shift. To maintain base class accuracy, some methods adopt strict parameter constraints, rendering them unable to effectively learn new knowledge; meanwhile, other methods suffer from feature extraction failures across multiple categories. In contrast, MLPPN alleviates this issue through effective feature decoupling. Under the two-shot setting, it not only achieves the highest accuracy among comparative methods on multiple fine-grained objects such as "airplane," "mobile home," and "pavement," but its mean IoU across categories also reaches 48.72%, demonstrating better category identifiability.

3) Results on the Vaihingen Dataset: To evaluate the performance of the proposed method on remote sensing

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TABLE VII
RESULTS OF EACH CATEGORY IN THE **TWO-SHOT** SETTING ON DLRSD DATASET (%)

Category	FT	WI	DWI	RT	AMP	LWF	ILT	MIB	PIFS	NEST	PLOP	MBS	GAPS	MLPPN
Airplane	9.96	13.91	15.33	0.00	38.00	5.34	23.65	15.59	0.00	39.07	59.21	0.00	47.66	**67.27**
Bare Soil	2.15	28.96	28.55	24.83	26.15	0.66	7.36	9.42	0.00	44.18	46.68	40.62	37.22	**48.84**
Building	1.17	50.23	50.69	8.03	59.45	9.10	11.48	23.03	0.00	59.42	**65.84**	49.10	56.98	55.39
Car	3.99	36.98	40.16	0.35	40.24	1.92	16.08	20.70	0.00	53.88	64.91	52.47	52.95	**68.01**
Shrub	0.00	30.92	27.78	27.82	1.42	0.00	7.71	0.06	0.00	32.25	51.08	36.43	46.29	**53.82**
Sports Field	0.00	54.39	54.42	0.57	71.25	5.10	11.64	0.46	0.00	69.17	**80.23**	66.58	64.21	77.89
Pier	7.09	0.00	0.00	0.15	0.03	11.45	12.51	2.88	0.00	0.00	13.79	0.00	**39.30**	0.00
Field	13.38	77.76	63.64	20.99	82.44	20.09	8.09	3.33	0.00	88.10	91.56	0.00	**96.45**	87.19
Grass	0.00	42.68	41.96	0.93	45.49	0.01	0.00	0.17	0.00	46.88	46.01	55.16	**61.49**	47.05
Mobile Home	0.00	44.18	43.02	0.00	51.15	0.17	0.00	6.33	0.00	57.47	67.30	45.43	50.38	**68.48**
Pavement	0.53	59.99	60.44	38.36	52.41	0.04	0.17	2.96	0.00	67.36	73.52	66.45	74.86	**76.08**
Beach	0.11	12.03	12.53	11.74	16.57	0.04	0.00	7.38	0.00	0.00	40.80	15.42	**71.89**	56.43
Sea	0.00	34.03	33.12	0.00	**35.59**	0.00	0.00	0.00	0.00	31.76	0.00	0.00	1.03	34.28
Ship	2.64	36.25	34.96	0.00	41.64	2.98	1.98	0.00	0.00	0.00	0.00	**47.92**	1.25	44.54
Storage Tank	0.00	3.03	3.56	0.00	3.95	0.00	0.00	0.00	0.00	6.27	0.00	25.83	**26.22**	7.68
Tree	0.00	16.30	7.54	2.04	2.25	2.28	0.01	0.10	0.00	6.51	0.00	0.00	**41.23**	8.81
Water Body	4.66	40.62	40.77	0.22	33.88	4.59	4.80	9.06	4.17	**41.95**	0.00	0.00	2.14	26.50
Mean	2.69	34.25	32.85	8.00	35.41	3.75	6.20	5.97	0.25	37.90	41.23	29.49	45.39	**48.72**

TABLE VIII
METRIC STATISTICS FOR **TWO-SHOT** ON THE VAIHINGEN DATASET (3–2)

Method	$mIoU_{base}$	$mIoU_{novel}$	$mIoU_{all}$	Recall	Precision
AMP	62.36	14.22	38.29	56.14	50.96
DWI	67.24	3.02	35.13	45.15	**56.94**
FT	58.54	7.38	32.96	41.34	49.67
ILT	58.25	7.65	32.95	41.25	49.96
LWF	59.39	7.16	33.27	41.45	51.27
MCLOS	9.68	0.00	4.84	12.50	4.84
MIB	59.94	8.53	34.23	42.75	49.67
PIFS	8.84	6.18	7.51	16.07	11.13
RT	56.22	10.46	33.34	43.11	48.04
WI	67.10	15.64	41.37	53.72	55.00
NEST	48.34	1.08	24.71	43.98	39.01
PLOP	**77.02**	0.00	38.51	39.98	47.38
MBS	70.75	0.03	35.39	40.81	44.76
GAPS	73.39	15.91	44.65	**71.33**	49.07
MLPPN	71.28	**20.23**	**45.76**	56.42	56.12

TABLE IX
METRIC STATISTICS FOR **FIVE-SHOT** ON THE VAIHINGEN DATASET (3–2)

Method	$mIoU_{base}$	$mIoU_{novel}$	$mIoU_{all}$	Recall	Precision
AMP	63.66	19.96	41.81	61.16	53.18
DWI	67.30	4.06	35.68	45.44	**57.52**
FT	64.13	2.04	33.09	42.20	52.77
ILT	64.02	1.88	32.95	42.08	52.41
LWF	65.02	3.18	34.10	43.26	53.62
MCLOS	7.37	0.00	3.69	13.85	7.96
MIB	65.22	4.58	34.90	44.04	51.22
PIFS	11.46	0.00	5.73	12.79	11.11
RT	63.53	6.59	35.06	44.13	48.39
WI	67.44	17.05	42.25	54.10	55.24
NEST	48.25	0.49	24.37	42.63	38.98
PLOP	**77.02**	0.00	38.51	39.98	47.38
MBS	70.87	0.08	35.47	41.04	44.93
GAPS	73.54	15.99	44.77	**71.61**	55.26
MLPPN	71.76	**20.46**	**46.11**	56.47	56.42

images with high resolution and complex spectral features, we conducted two-shot and five-shot few-shot incremental semantic segmentation experiments under the 3–2 setting on the ISPRS Vaihingen dataset. The quantitative results (Tables VIII and IX) indicate that MLPPN exhibits strong competitiveness in overall segmentation performance. Under the two-shot and five-shot settings, the overall mean intersection over union ($mIoU_{all}$) of MLPPN reaches 45.76% and 46.11%, respectively, both representing the highest levels among all comparative methods. Particularly in terms of the learning capability for novel categories ($mIoU_{novel}$), MLPPN achieves accuracies of 20.23% and 20.46% under the two settings, respectively, significantly outperforming runner-up models such as AMP, WI, and GAPS. Observing the data, it can be found that although some baseline methods (such as PLOP and MBS) maintain a high accuracy on base classes ($mIoU_{base}$) (e.g., 77.02% for PLOP), their novel class accuracy is close to or even equal to 0.00%. This indicates that these methods adopt an extremely conservative strategy when dealing with complex land covers, failing to effectively establish feature representations for novel classes. In contrast, MLPPN achieves a more reasonable balance between learning new land cover features and retaining old class knowledge.

Combining the statistical metrics of recall and precision, MLPPN also demonstrates a relatively stable comprehensive prediction capability. Under the two-shot and five-shot settings, the precision of MLPPN reaches 56.12% and 56.42%, respectively, securing the highest values among all comparative methods. This demonstrates that the proposed method can effectively reduce false-positive predictions for complex backgrounds or easily confused land covers, ensuring the output positive samples are more reliable. It is worth noting that although methods like GAPS achieve higher values in recall (e.g., 71.33% under the two-shot setting), their precision is relatively low (49.07%), meaning that their higher recall is often accompanied by an excessively loose classification of pixels. Meanwhile, MLPPN maintains a recall rate of over 56% while keeping the highest precision, reflecting that the

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TABLE X
METRIC STATISTICS FOR **TWO-SHOT** ON THE POTSDAM DATASET (3–2)

Method	$mIoU_{base}$	$mIoU_{novel}$	$mIoU_{all}$	Recall	Precision
AMP	50.77	**24.07**	37.42	51.45	47.38
DWI	56.69	7.31	32.00	48.08	54.30
FT	51.67	15.84	33.76	46.44	50.12
ILT	51.67	17.25	34.46	47.25	49.81
LWF	53.22	17.85	35.53	48.71	50.13
MCLOS	6.83	0.00	3.42	12.50	3.42
MIB	48.02	8.27	28.14	41.85	44.14
PIFS	3.24	14.43	8.84	22.68	23.40
RT	38.90	15.58	27.24	40.14	46.44
WI	44.58	20.10	32.34	51.10	**60.21**
NEST	40.74	18.72	29.73	49.18	45.78
PLOP	26.72	0.00	13.36	25.25	25.84
MBS	53.55	0.00	26.77	33.52	38.66
GAPS	**61.14**	3.60	32.37	**57.76**	43.60
MLPPN	58.21	19.20	**38.71**	52.62	59.84

TABLE XI
METRIC STATISTICS FOR **FIVE-SHOT** ON THE POTSDAM DATASET (3–2)

Method	$mIoU_{base}$	$mIoU_{novel}$	$mIoU_{all}$	Recall	Precision
AMP	50.53	20.15	35.34	54.66	52.77
DWI	60.36	15.86	38.11	47.48	60.44
FT	57.48	19.47	38.48	50.96	58.02
ILT	57.22	19.57	38.39	50.84	58.20
LWF	60.21	18.60	39.41	54.64	58.77
MCLOS	6.83	0.00	3.42	12.50	3.42
MIB	53.52	6.14	29.83	44.91	51.51
PIFS	1.48	11.90	6.69	24.54	10.73
RT	50.34	19.14	34.74	47.03	55.40
WI	46.42	21.49	33.95	53.22	60.71
NEST	57.11	**21.83**	39.47	50.04	52.14
PLOP	34.67	0.46	17.56	38.70	31.46
MBS	53.69	0.08	26.88	33.74	38.83
GAPS	**61.25**	3.75	32.50	**58.07**	43.79
MLPPN	57.46	21.67	**39.56**	53.63	**63.11**

model has reached a better optimized state between mining complete regions of land covers and controlling classification errors.

4) Results on the Potsdam Dataset: To evaluate the generalization and continual learning capabilities of the model on high-resolution remote sensing images with complex distributions of urban objects, we conducted two-shot and five-shot few-shot incremental semantic segmentation experiments under the 3–2 setting on the Potsdam dataset. The quantitative results (Tables X and XI) indicate that MLPPN still maintains a leading edge in overall segmentation performance. Under the two-shot and five-shot settings, the overall mean intersection over union ($mIoU_{all}$) of MLPPN reaches 38.71% and 39.56%, respectively, achieving the highest values among all compared methods. Observing the specific category accuracy metrics, it can be found that although some baseline methods (such as GAPS) achieve the highest accuracy on base classes ($mIoU_{base}$) with 61.14% (two-shot) and 61.25% (five-shot), their novel class accuracy ($mIoU_{novel}$) is only 3.60% and 3.75%, respectively, indicating a failure to effectively learn novel categories while retaining old knowledge. Conversely, AMP shows higher novel class accuracy under the two-shot setting (24.07%), but its base class accuracy and overall performance are inferior to MLPPN. Similarly, the novel class accuracy of methods like PLOP and MBS is close to or equal to 0.00%, making it difficult to effectively extract features of novel objects. MLPPN achieves novel class accuracies of 19.20% and 21.67% under two-shot and five-shot settings, respectively, effectively learning new knowledge while better maintaining base class accuracy, realizing a more reasonable balance between stability and plasticity.

Further analysis combining recall and precision statistics reveals that MLPPN exhibits a relatively stable comprehensive prediction level in complex urban scenes. Under the five-shot setting, the precision of MLPPN reaches 63.11%, the highest among all comparative methods. Under the two-shot setting, its precision (59.84%) is also only second to WI (60.21%), placing it in the leading echelon. Although GAPS achieves the highest recall under both settings (57.76% and 58.07%, respectively), its precision is relatively low (43.60% and 43.79%). This indicates that the higher recall of GAPS is traded off through overly broad pixel partitioning, which tends to introduce more false-positive predictions. In contrast, MLPPN realizes higher precision while maintaining a reasonable recall rate (52.62% and 53.63%), indicating that its output segmentation masks are more reliable in reducing local misclassifications.

D. Qualitative Analysis

For clarity, compact color legends are provided in the qualitative result figures for the corresponding datasets, and the captions indicate the incremental novel classes in the shown settings. Specifically, the novel classes are structure/vegetation/road/sky for aeroscapes, sea/ship/storage tank/tree/water body for DLRSD, vehicle/architect for Vaihingen, and tree/car for Potsdam.

It should be noted that the qualitative visualizations are not intended to show only successful cases. They are used to present representative incremental settings and to intuitively analyze how the multilevel feature representation and prototype-based matching mechanism behave in complex scenes. Although the visual results demonstrate the relative advantages of MLPPN, they also reveal several remaining limitations. In complex backgrounds, MLPPN may still generate local noisy predictions and fragmented boundaries. For visually similar categories, such as tree and low vegetation in high-resolution urban scenes, the model may confuse local regions when the support samples do not sufficiently cover intraclass variations. Small objects are also challenging because only a limited number of annotated pixels are available for constructing stable prototypical parts. These observations indicate that MLPPN alleviates, but does not completely remove, the difficulty of few-shot incremental segmentation in complex remote sensing scenes.

1) Qualitative Analysis on Aeroscapes Dataset: From the qualitative comparison results shown in Fig. 3, it is evident that the few-shot incremental segmentation task poses significant challenges on the aeroscapes dataset, which features complex backgrounds and large-scale drone aerial perspectives. Observing the majority of baseline methods, they exhibit distinct vulnerability when confronted with the impact of new

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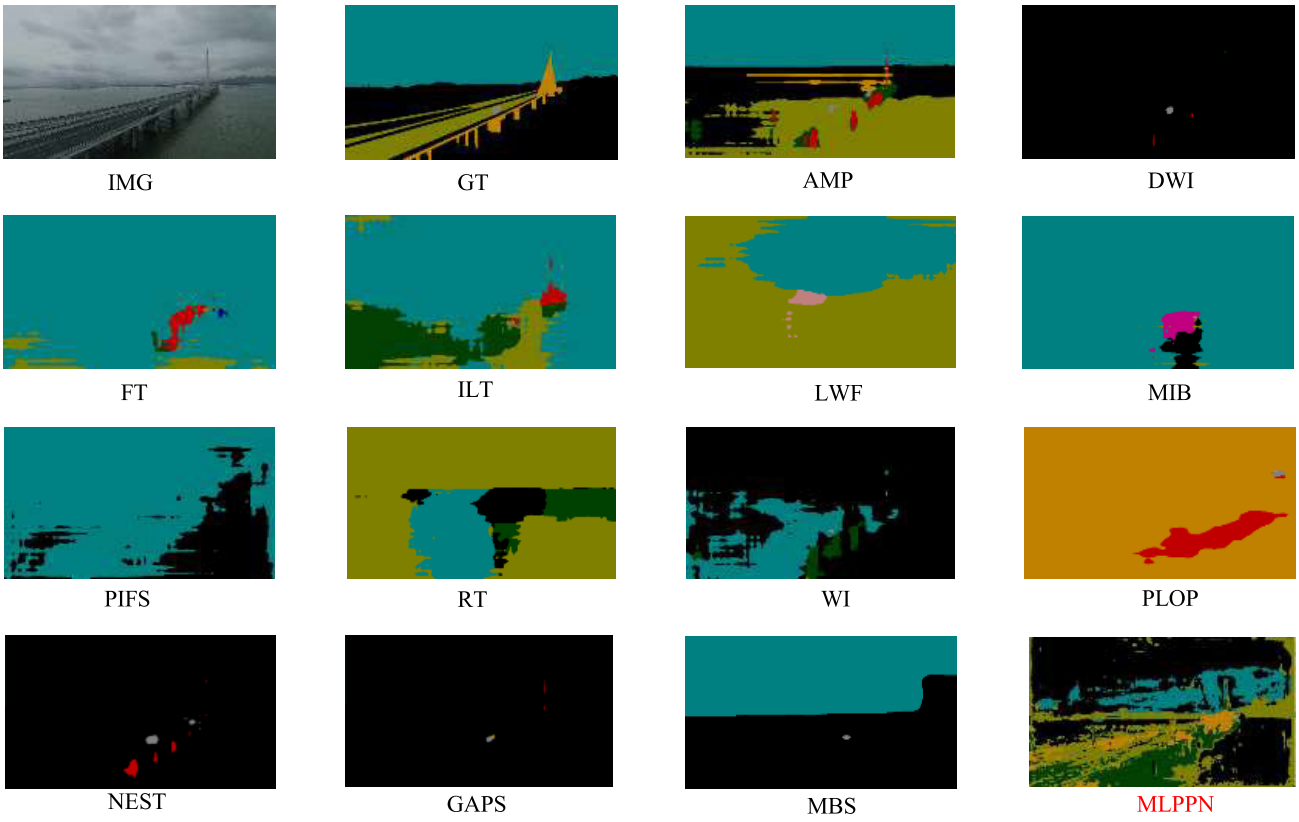


Fig. 3. Qualitative comparison of segmentation results under a two-shot setting on the aeroscapes dataset. "IMG" denotes the input image, and "GT" represents the ground truth. Compared to various baseline methods, our proposed MLPPN effectively mitigates catastrophic forgetting and demonstrates superior structural preservation. It yields the most visually coherent segmentation mask that closely resembles the GT, accurately distinguishing complex regions like sky, water, and the bridge.

knowledge. For instance, the prediction results of methods such as DWI, NEST, GAPS, and MBS appear almost entirely black with only sporadic noise remaining. This indicates that these models suffer from severe catastrophic forgetting during the incremental learning phase, completely losing their discriminative ability for base classes (e.g., sky and water). Conversely, other methods (such as PLOP, FT, and LWF) retain responses to certain features but suffer from severe semantic confusion and structural collapse. They generate a large number of erroneous patches that contradict the real scene, completely failing to maintain the global consistency and object boundaries of the image.

In contrast, our proposed MLPPN method demonstrates remarkable superiority and robustness in visual presentation. As shown in Fig. 3, the segmentation mask generated by MLPPN is the closest to the GT. It not only successfully overcomes the catastrophic forgetting problem prevalent in other methods but also exhibits outstanding structural preservation capabilities. When dealing with complex aerial scenes, MLPPN can clearly and coherently delineate the boundaries of the sky, the water background, and the bridge structures cutting across the image. This fully proves that while learning novel categories relying on only a few samples, our method effectively protects the model's understanding of prior knowledge and global context. It achieves an ideal balance between plasticity and stability in incremental learning, ultimately outputting high-quality and highly coherent segmentation results.

2) Qualitative Analysis on DLRSD Dataset: The qualitative visualization results intuitively demonstrate the actual segmentation performance of the models across continuous incremental stages. After undergoing one incremental session in a complex urban crossroad scenario (Fig. 4), most baseline methods exhibit feature confusion when new categories are introduced, failing to effectively distinguish target objects from background regions, which leads to predictions converging to a single category. In contrast, MLPPN better maintains the spatial topology of the road and surrounding objects in the early incremental stages, and its semantic boundaries are closer to the GT. After undergoing five complete incremental stages (Fig. 5), it can be observed that large cyan color blocks appear in the prediction maps of some baseline models (e.g., RT, ILT, LWF, and PIFS). Since these models were just trained on the "water body" (corresponding to cyan) category in the fifth incremental stage, this phenomenon indicates that they suffer from severe overfitting to the most recently learned category. Therefore, when testing on this image that originally contains no water bodies, they erroneously classify a large number of background or road pixels as water. Conversely, MLPPN's prediction results do not show such large-scale specific-class misclassifications, indicating that the model better overcomes the overfitting problem to the latest task. Although MLPPN's predictions in the final stage also contain some local noise, it overall retains the spatial distribution features of the main objects (such as the road network), demonstrating a more objective and stable continual segmentation capability.

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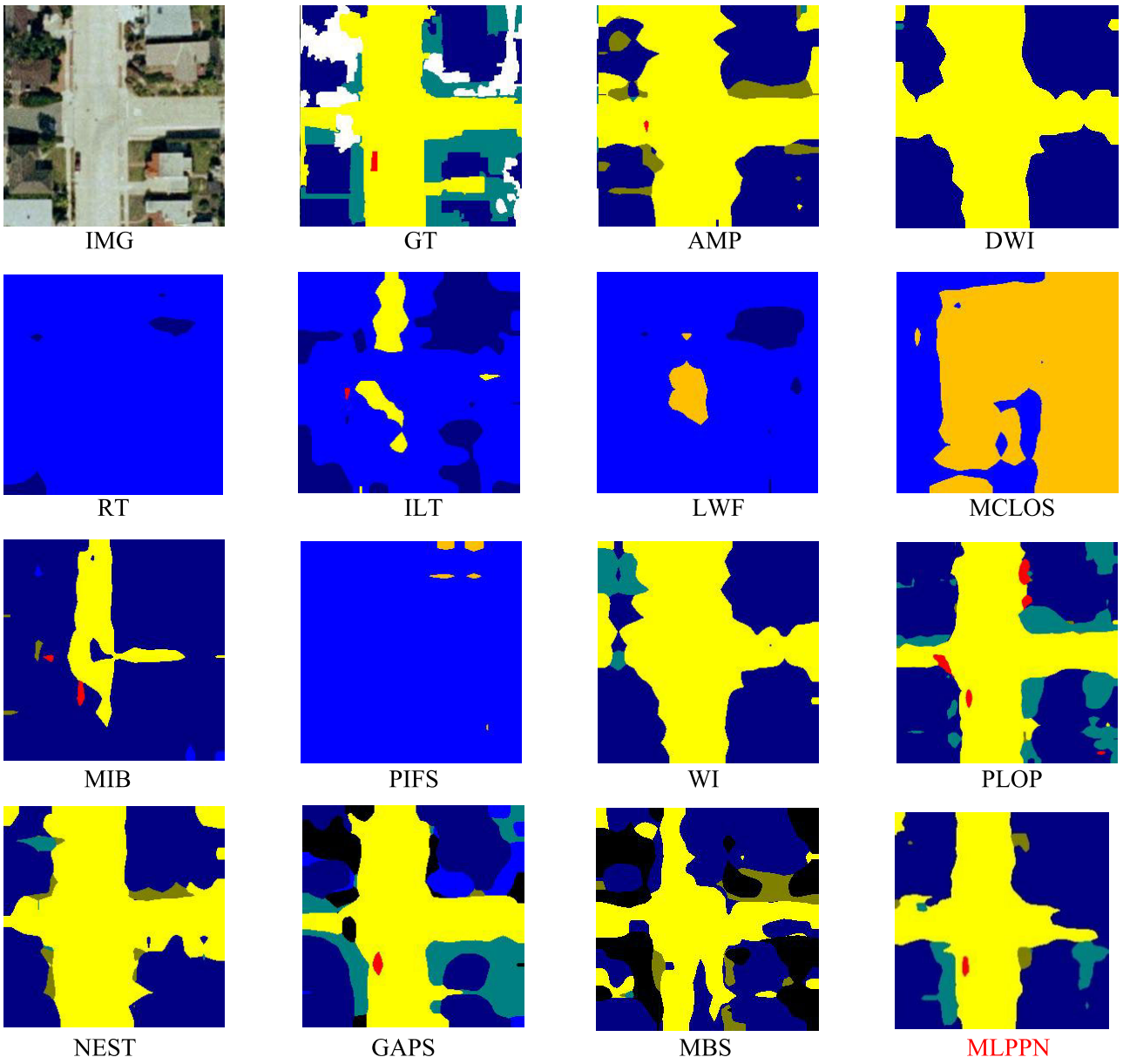


Fig. 4. Qualitative comparison of few-shot incremental segmentation results after one incremental stage under the two-shot setting on the DLRSD dataset. While some baseline methods lose discriminative ability and others produce fragmented masks, our proposed MLPPN successfully maintains the structural integrity of the road network (yellow) and surrounding buildings (dark blue), closely aligning with the GT.

3) Qualitative Analysis on Vaihingen Dataset: The qualitative visualization results (Fig. 6) further verify the classification capability of the model in actual complex remote sensing scenes. The Vaihingen dataset contains a large number of vegetation categories with extremely similar spectral features (such as "tree" and "low vegetation"), which significantly increases the classification difficulty during the incremental stages. From the visualized segmentation maps under the two-shot setting, it can be seen that most traditional baseline methods (such as PLOP, MIB, FT, and LWF) exhibit obvious category confusion when distinguishing these two types of vegetation. They generally fail to effectively extract the differential features between old and new vegetation, causing the prediction results to largely converge to a single category (e.g., predicting almost all vegetation areas as the cyan "low vegetation"), thereby losing the ability to recognize the green "tree" class. Some methods (such as PIFS and NEST) even show large-area background misclassifications or semantic category errors. Under the same conditions, MLPPN can relatively accurately distinguish these spectral categories with high similarity, better restoring the actual spatial distribution of different vegetation types. Furthermore, MLPPN also preserves relatively clear and sharp boundaries for artificial objects such as buildings. Although there are still some localized sporadic noises in the prediction maps, it overall proves that MLPPN has a superior objective performance in mitigating forgetting and distinguishing confusing categories.

4) Qualitative Analysis on Potsdam Dataset: Qualitative visualization results further intuitively reflect the actual segmentation capabilities of each model on the Potsdam dataset. Fig. 7 displays the segmentation results for complex urban objects, including buildings (red), cars (cyan), and vegetation

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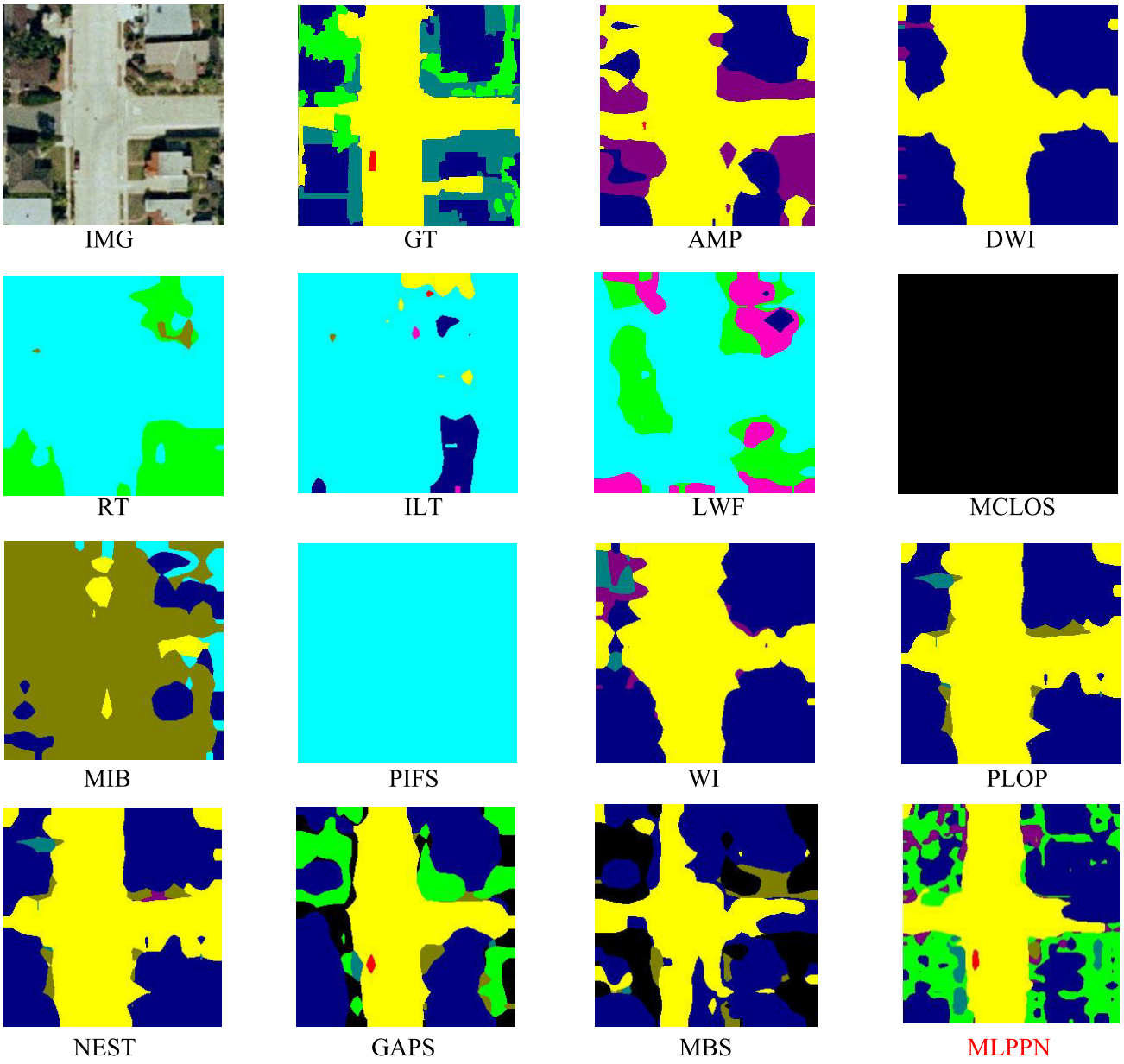


Fig. 5. Qualitative comparison of few-shot incremental segmentation results after five incremental stages under the two-shot setting on the DLRSD dataset. After continuous incremental learning, several baseline methods exhibit severe overfitting to the newly learned "water" class in the final stage, falsely predicting massive light blue areas despite the absence of water in the actual image. In contrast, MLPPN effectively prevents this catastrophic overfitting and forgetting, robustly preserving the accurate spatial structures of previously learned classes.

(green), under the five-shot setting. It can be seen that due to the dense spatial distribution and complex features of urban objects, several baseline models suffer from obvious category confusion or prediction homogenization. For example, MBS erroneously predicts the entire image as a single building category (red), failing to distinguish other objects. PIFS loses basic spatial structural representation, yielding output results with irregular color block distributions. In addition, methods like GAPS and PLOP fail to effectively recognize the vegetation (green) regions in the image. Under the same experimental conditions, MLPPN relatively completely preserves the main outlines of the red buildings and successfully locates the cyan car targets as well as the green vegetation regions. Although MLPPN has some fragmentation and local noise in predicting the edges of vegetation and backgrounds, overall, it effectively avoids the large-scale misclassifications and category homogenization commonly seen in baseline models, demonstrating a relatively stable analytical capability for complex scenes.

E. Ablation Experiments

Table XII presents the ablation study results on the DLRSD dataset under the five-shot setting regarding the feature layers for prototype generation and the number of prototypical parts in each layer. By adjusting the combination of feature extraction layers (a–b–c) from the backbone network and the corresponding number of clustered parts (x–y–z), the

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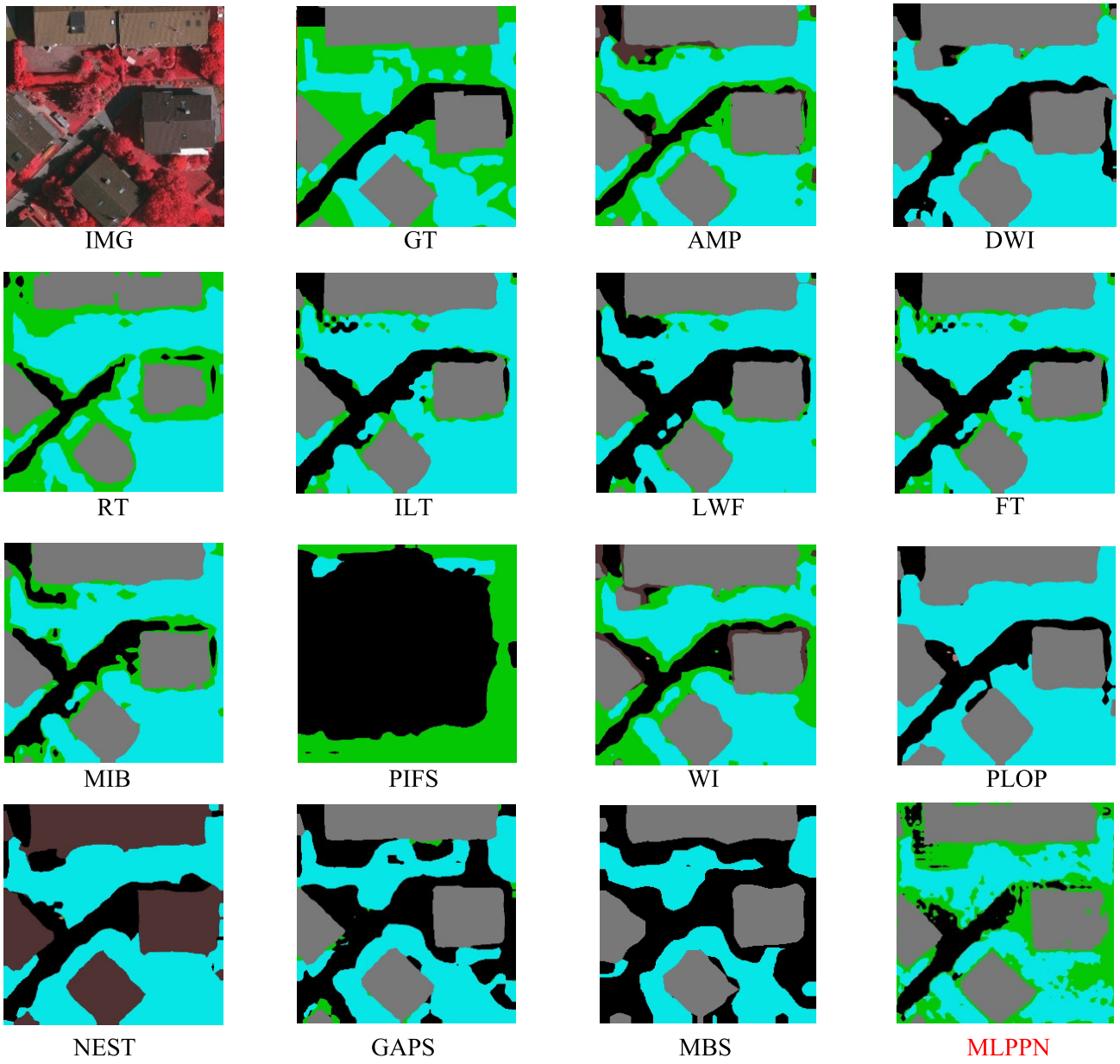


Fig. 6. Qualitative segmentation results under a two-shot setting on the Vaihingen dataset. Traditional methods exhibit severe catastrophic forgetting or interclass confusion between similar vegetation types. MLPPN effectively distinguishes highly similar spectral classes and preserves sharp boundaries without forgetting.

experiment delves into the profound impact of multiscale features and part-level decoupling on incremental segmentation performance.

First, regarding the selection of feature layers, comparing the optimal configuration (two- to four-neck) with other variants reveals that introducing excessively shallow features (e.g., the (one- to four-neck) configuration) brings in a massive amount of low-level detail noise. This leads to a chaotic metric space for novel class features, causing a sharp decline in the novel class mean intersection over union ($mIoU_{novel}$) across all stages. Conversely, relying solely on higher level features [e.g., the (three- and four-neck) or (one-neck) configurations] causes the model to lose critical mid-level texture and structural details, rendering it similarly unable to effectively represent complex new land covers. Combining the mid-level features of Layer 2 with high-level semantics provides the most discriminative multiscale feature representation.

Second, in the discussion of the number of prototypical parts, the (three- and four-neck)-(1–1–1) setting represents the traditional strategy of generating only a single global mean prototype for each category. Although this conservative strategy exhibits high stability in base class accuracy ($mIoU_{base}$), its novel class performance and overall prediction precision fall far short of the (2–3–3) configuration. This compellingly proves that a single prototype feature fundamentally cannot encompass the extremely high physical discrepancies and morphological intraclass variance inherent in remote sensing land covers. The (3–3–3) multipart decoupling strategy ultimately

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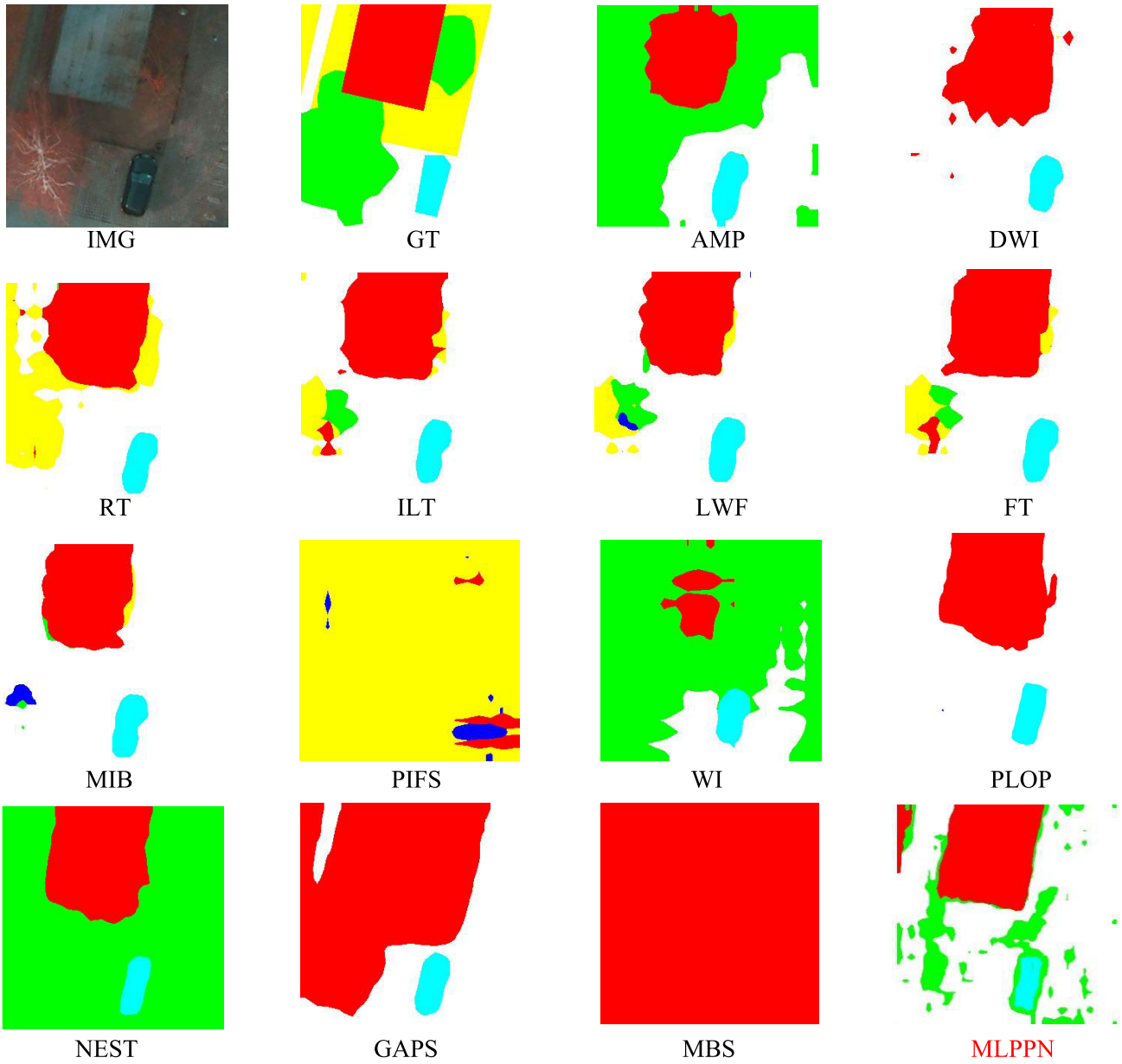


Fig. 7. Qualitative segmentation results under a five-shot setting on the Potsdam dataset.

adopted in this article successfully breaks down complex land covers into diverse visual primitives. With almost no sacrifice to base class accuracy, it achieves a substantial enhancement in novel class representational capacity, enabling the model to reach the most ideal balance in comprehensive performance metrics (overall mean intersection over union, precision, and recall) across the five lengthy incremental stages.

Table XIII presents the ablation study results of the attention mechanisms. To scientifically evaluate the balance between old and new knowledge, this article adopts the harmonic mean (HM_{IoU}) as the core evaluation metric for intersection over union, while recall and precision are computed using the standard macro-average over all categories.

Observing the experimental data reveals a typical phenomenon of "false high recall." Although the Baseline model (without any attention) achieves the highest recall (73.76%) in the first stage, this is attributed to its chaotic feature boundaries, which forcibly increase the hit rate through massive over-prediction of novel categories. This strategy leads to an enormous amount of false positives, restricting its precision to a mere 48.13% and severely limiting its overall HM_{IoU}. Conversely, the variant using only intralayer attention falls into an extremely conservative state; it is overly reluctant to predict novel objects, resulting in a precipitous drop in recall.

The complete architecture (full model) finally constructed in this article perfectly resolves this extreme contradiction through the synergistic dynamic weighting of interlayer and intralayer attention. The full model not only achieves an absolute lead in HM_{IoU} across all stages (reaching 42.19% in S1) but also surpasses all comparison variants in

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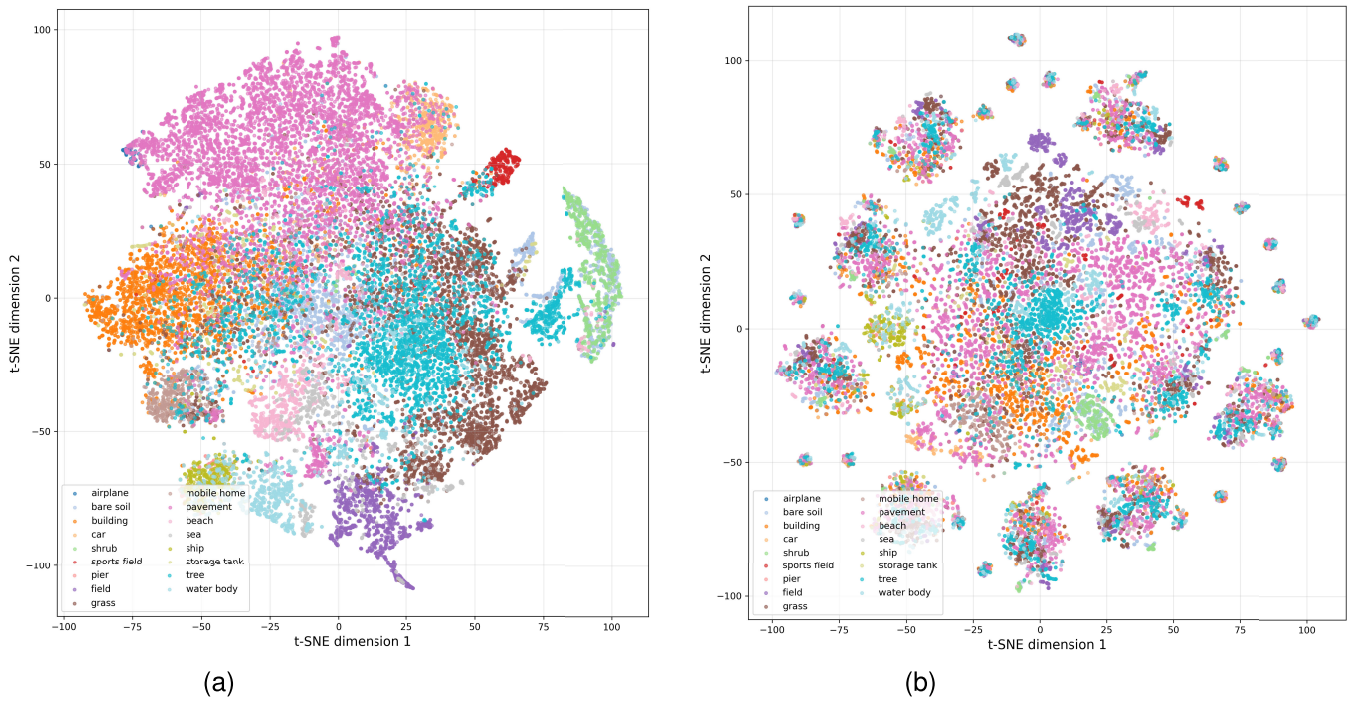


Fig. 8. t-SNE visualization of the high-level features (Layer 4) on the DLRSD dataset at the final incremental stage. (a) Highly structured and coherent manifold of the proposed MLPPN, while (b) shows significant feature entanglement and fragmentation in the baseline method.

TABLE XII

ABLATION STUDY ON PARAMETERS FOR **FIVE-SHOT** ON THE **DLRSD** DATASET. (a–b–c)–(x–y–z) INDICATES THAT PROTOTYPES ARE GENERATED IN LAYERS a, b, AND c, WITH THE NUMBER OF PROTOTYPICAL PARTS IN EACH LAYER BEING x, y, AND z, RESPECTIVELY

Settings	Step	$mIoU_{base}$	$mIoU_{novel}$	$mIoU_{all}$	Recall	Precision
(1-4-neck)-(2-3-3)	S1	66.54	7.93	37.24	42.70	71.70
	S2	59.05	37.74	48.40	61.59	68.22
	S3	58.41	25.53	41.97	53.37	57.40
	S4	54.82	12.90	33.86	48.59	44.85
	S5	54.37	16.10	35.23	49.33	47.87
(2-4-neck)-(1-1-1)	S1	**68.45**	38.17	53.31	60.47	77.80
	S2	**60.82**	53.91	57.37	70.89	71.96
	S3	**59.99**	37.62	**48.80**	61.66	60.48
	S4	52.07	**28.63**	40.35	53.42	51.22
	S5	51.62	**25.72**	**38.67**	52.54	53.92
(2-4-neck)-(2-3-3)	S1	68.16	**44.77**	**56.46**	**63.01**	**82.47**
	S2	60.60	**55.11**	**57.86**	**72.14**	**76.54**
	S3	58.88	**38.52**	48.70	**64.04**	**64.76**
	S4	54.95	27.26	**41.10**	**56.60**	**57.38**
	S5	54.36	22.37	38.36	**54.78**	**59.41**
(3-4-neck)-(2-3-3)	S1	67.16	18.27	42.72	49.89	63.95
	S2	59.61	42.21	50.91	65.11	63.73
	S3	59.04	30.13	44.59	60.03	57.34
	S4	**55.42**	20.17	37.80	53.70	55.41
	S5	**54.84**	16.27	35.56	52.22	55.86
(4-neck)-(2-3)	S1	67.26	9.14	38.20	45.26	50.74
	S2	59.67	39.17	49.42	63.23	57.95
	S3	59.01	27.51	43.26	56.17	50.95
	S4	54.90	16.66	35.78	50.79	42.61
	S5	54.43	13.55	33.99	49.83	45.90

prediction precision. This compellingly corroborates that the proposed dual attention mechanism effectively filters out noise interference and false alarms during the incremental process. It establishes highly discriminative decision boundaries, achieving the optimal balance between stability and plasticity without blindly sacrificing precision.

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TABLE XIII

ABLATION STUDY ON ATTENTION MECHANISMS FOR **FIVE-SHOT** ON THE **AEROSCAPES** DATASET. HM_{IoU} DENOTES THE HARMONIC MEAN OF $mIoU_{base}$ AND $mIoU_{novel}$. RECALL AND PRECISION ARE COMPUTED USING THE STANDARD MACRO-AVERAGE OVER BASE AND NOVEL CATEGORIES

Settings	Step	$mIoU_{base}$	$mIoU_{novel}$	HM_{IoU}	Recall	Precision
Baseline (w/o Attn)	S1	42.69	22.48	29.45	**73.76**	48.13
	S2	42.08	14.73	21.82	50.68	56.97
	S3	32.23	16.54	21.86	**46.26**	43.85
	S4	31.59	16.07	21.30	44.17	42.39
+ Intra-layer Attn	S1	**51.84**	11.45	18.76	38.08	58.47
	S2	**45.79**	10.75	17.41	37.65	60.76
	S3	**35.84**	7.01	11.73	35.59	46.91
	S4	**35.19**	5.16	9.00	34.43	40.61
+ Inter-layer Attn	S1	48.60	27.67	35.26	64.06	53.76
	S2	45.27	32.90	38.11	**53.47**	57.40
	S3	34.81	22.62	27.42	45.47	44.01
	S4	34.55	18.12	23.77	**46.24**	42.20
Full Model	S1	44.60	**40.03**	**42.19**	61.79	**63.50**
	S2	41.18	**36.59**	**38.75**	46.30	**65.77**
	S3	30.77	**27.55**	**29.07**	46.15	**51.10**
	S4	30.64	**24.09**	**26.98**	40.16	**48.33**

TABLE XIV

COMPARISON OF INFERENCE SPEED (FPS) ACROSS DIFFERENT METHODS ON THE DLRSD DATASET

Method	Incremental Step						Average
	S0	S1	S2	S3	S4	S5	
AMP	**38.37**	41.22	**41.90**	**42.11**	**42.01**	**41.66**	**41.21**
WI	38.33	**41.97**	41.48	41.34	41.37	41.26	40.96
NEST	36.45	36.20	36.11	36.43	35.98	36.15	36.22
MLPPN	31.30	32.83	31.88	32.88	32.06	31.90	32.14

TABLE XV

PROTOTYPE BANK SIZE AND MEMORY USAGE AT THE FINAL INCREMENTAL STAGE

Dataset	Classes	Prototypes	Memory
Aeroscapes	12	132	33.8 KB
DLRSD	18	198	50.7 KB
Vaihingen	6	66	16.9 KB
Potsdam	6	66	16.9 KB

F. Inference Speed Analysis

To evaluate the computational efficiency and practical applicability of our proposed method, we compare the inference speed of MLPPN with several representative continuous learning approaches on the DLRSD dataset. Table XIV reports the average processing speed, measured in frames per second (FPS), across six incremental stages. Each stage was executed three times, and the average FPS is presented.

As shown in the table, from the quantitative results, the proposed MLPPN achieves an average inference speed of 32.14 FPS, which is slightly lower than baseline methods such as AMP and WI. This is primarily because MLPPN needs to calculate the similarity between current features and stored category prototypes during the inference phase; this matching process introduces some additional computational overhead.

We further report the size of the prototype bank in Table XV. In MLPPN, each class stores one class-level mean prototype and multiple part-level prototypes at each selected feature level. With the adopted setting of $[2, 3, 3]$ prototypical parts across three feature levels, each class contains $(2 + 1) + (3 + 1) + (3 + 1) = 11$ stored prototypes. The projected prototype dimension is 64, and the prototypes are stored in 32-bit floating-point format. Therefore, even at the final incremental stage, the prototype bank requires only 33.8 KB on aeroscapes, 50.7 KB on DLRSD, and 16.9 KB on Vaihingen and Potsdam, which is negligible compared with the memory consumed by the backbone and intermediate feature maps.

Let L be the number of selected feature levels, let H_l and W_l be the spatial size of the feature map at level l, let C be the projected feature dimension, let N_t be the number of seen classes at incremental step t, and let P_l be the number of prototypes per class at level l. The matching complexity of the prototype bank is

$$O\left(\sum_{l=1}^{L} H_l W_l C N_t P_l\right) \tag{18}$$

which grows linearly with the number of seen classes. This explains why MLPPN introduces additional matching overhead but still maintains stable real-time inference speed across incremental stages.

Nevertheless, the experimental results also reveal two positive aspects. First, as the incremental steps progress (from Step 0 to Step 5) and the number of known categories increases, the inference speed of MLPPN remains stable between 31.30 and 32.88 FPS, showing no linear downward trend with the growing number of classes. This indicates that the prototype matching mechanism does not impose a continuously expanding computational burden, allowing the model to maintain consistent inference efficiency in long-sequence tasks. Second, when evaluating the overall time cost of incremental learning comprehensively, the slight latency in the inference stage is an acceptable tradeoff. Most existing baseline methods typically require time-consuming model

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FT or parameter updates when introducing new categories. In contrast, MLPPN is training-free during the incremental phase. The substantial time saved by omitting backpropagation largely offsets the minor loss in frame rate during inference. Overall, while maintaining a reasonable real-time performance (>30 FPS), MLPPN demonstrates the capability to rapidly respond to and process novel classes, offering an efficient and feasible solution for dynamic deployment in practical applications.

G. Visualization Analysis of the Feature Space

To more intuitively reveal the protective mechanism over the feature manifold and the representational superiority of the proposed method during the incremental learning process, high-level features (Layer 4) from the test set in the fifth incremental stage on the DLRSD dataset were extracted. The t-SNE algorithm is then utilized to conduct dimensionality reduction and visualization comparison of the feature spaces between the proposed method and the comparison method.

As shown in Fig. 8, observing the feature distribution of the baseline method, it is evident that its feature space exhibits obvious symptoms of catastrophic forgetting and representation collapse. In the central region of the 2-D plane, massive feature points from different categories suffer from severe feature entanglement, forming an unresolvable mixed cluster. This indicates that the model has completely lost its discriminative capacity for these categories. Simultaneously, a large number of abnormally fragmented and isolated tiny clusters appear at the periphery of its feature space. This demonstrates that traditional gradient-based incremental updates, when adapting to new samples, destructively force the reshaping of network parameters, completely shattering the original feature manifold of the old categories.

In stark contrast, the proposed MLPPN exhibits a highly structured and coherent feature distribution. Due to the adoption of the absolute parameter freezing strategy and the tuning-free prototypical imprinting mechanism, no abnormal feature fragmentation phenomenon occurs within the feature space of MLPPN. More importantly, despite the extreme complexity of pixel-level features in remote sensing images, the proposed method still demonstrates excellent intraclass compactness and interclass separability. Feature points belonging to the same category (e.g., Classes 3, 6, and 12) spontaneously aggregate within their exclusive feature neighborhoods, forming relatively clear and smooth decision boundaries between different categories. This visualization result compellingly corroborates from the deep feature level that the multilevel prototypical part decoupling mechanism based on cognitive assimilation can construct an extremely stable and highly discriminative metric space for the model, thereby fundamentally resisting feature degradation during the few-shot incremental process.

V. Conclusion

This article addresses the overfitting and catastrophic forgetting challenges in few-shot incremental land-cover segmentation by proposing a tuning-free MLPPN, inspired by the RBCs cognitive theory. By freezing model parameters after base training and using gradient-free feature clustering for novel-class registration, MLPPN decouples base representation learning from incremental prototype construction and alleviates the memory-forgetting drawbacks of traditional FT paradigms. To tackle the high intraclass variance of complex remote sensing composites, the model constructs a multiscale part-level representation space by leveraging both mid-level textural details and high-level semantics. Optimized via prototype-separation losses and guided by multilevel adaptive attention alongside an unbiased inference strategy, MLPPN improves the discriminability of confusing land covers. Extensive experiments on four mainstream benchmarks, including the aeroscapes, DLRSD, and ISPRS Vaihingen datasets, confirm that the proposed method achieves competitive performance under challenging two-shot and five-shot settings. Despite these encouraging results, MLPPN still has several limitations. First, the dense matching between query features and the prototype bank introduces additional inference cost compared with a standard parametric classifier. Second, local noise and fragmented predictions may still appear in scenes with complex backgrounds, small objects, or highly similar land-cover categories. Third, as the number of incremental sessions continues to grow, the prototype bank may become increasingly large, making prototype compression, selection, and long-term management important directions for future work.

References

[1] L. Huang, B. Jiang, S. Lv, Y. Liu, and Y. Fu, "Deep-learning-based semantic segmentation of remote sensing images: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8370–8396, 2024.

[2] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[3] B. Ren, Q. Wang, B. Liu, B. Hou, C. Yang, and L. Jiao, "WHFNet: A wavelet-driven heterogeneous fusion network for high-frequency enhanced optical–SAR remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5607817.

[4] B. Ren et al., "Incremental land cover classification via soft label and subregion distillation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5647322.

[5] B. Ren et al., "AFDINet: Adaptive frequency disentanglement and interaction in hybrid mamba–CNN for optical-SAR land cover classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 151, Jul. 2026, Art. no. 105406.

[6] D.-W. Zhou, Q.-W. Wang, Z.-H. Qi, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Class-incremental learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9851–9873, Dec. 2024.

[7] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, May 2021.

[8] J. Zeng, Y. Tan, L. Yang, S. Zhang, and L. Liang, "TerraSAP: Spatially aware prompt-based framework for few-shot class-incremental learning in remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 3143–3156, 2025.

[9] H. Liu et al., "Few-shot class-incremental learning via entropy-regularized data-free replay," in *Computer Vision–*(ECCV), S. Avidan, G. Brostow, M. Cissé, G. M. Farinella, and T. Hassner, Eds., Cham, Switzerland: Springer, 2022, pp. 146–162.

[10] M. McCloskey and N. J. Cohen, "*Catastrophic interference in connectionist networks: The sequential learning problem*," in *Psychology of Learning and Motivation*, vol. 24. Amsterdam, The Netherlands: Elsevier, 1989, pp. 109–165.

[11] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.

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[12] N. Dong and E. P. Xing, "Few-shot semantic segmentation with prototype learning," in *Proc. BMVC*, 2018, vol. 3, no. 4, p. 4.
[13] G. Li, V. Jampani, L. Sevilla-Lara, D. Sun, J. Kim, and J. Kim, "Adaptive prototype learning and allocation for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8330–8339.
[14] Y. Liu, X. Zhang, S. Zhang, and X. He, "Part-aware prototype network for few-shot semantic segmentation," 2020, *arXiv:2007.06309*.
[15] I. Biederman, "Recognition-by-components: A theory of human image understanding," *Psychol. Rev.*, vol. 94, no. 2, pp. 115–147, 1987.
[16] Y. Zou, S. Zhang, J. Yu, Y. Tian, and J. M. F. Moura, "Revisiting mid-level patterns for cross-domain few-shot recognition," 2020, *arXiv:2008.03128*.
[17] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–22.
[18] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[19] O. Ronneberger et al., "*U-Net: Convolutional networks for biomedical image segmentation*," in *Medical Image Computing and Computer-Assisted Intervention–(MICCAI)*. Cham, Switzerland: Springer, 2015, pp. 234–241.
[20] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," 2020, *arXiv:2002.05709*.
[21] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[22] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," 2018, *arXiv:1802.02611*.
[23] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.
[24] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.
[25] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.
[26] G. M. van de Ven, T. Tuytelaars, and A. S. Tolias, "Three types of incremental learning," *Nature Mach. Intell.*, vol. 4, no. 12, pp. 1185–1197, Dec. 2022.
[27] G. Saha and K. Roy, "Saliency guided experience packing for replay in continual learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 5262–5272.
[28] J. Xie, B. Pan, X. Xu, and Z. Shi, "MiSSNet: Memory-inspired semantic segmentation augmentation network for class-incremental learning in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607913.
[29] Y. Guo et al., "Decoupling continual semantic segmentation," 2025, *arXiv:2508.05065*.
[30] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.
[31] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[32] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," 2017, *arXiv:1706.08840*.
[33] A. Chaudhry et al., "On tiny episodic memories in continual learning," 2019, *arXiv:1902.10486*.
[34] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," 2017, *arXiv:1703.03400*.
[35] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," 2018, *arXiv:1812.00420*.
[36] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," 2017, *arXiv:1703.04200*.
[37] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," 2016, *arXiv:1611.07725*.
[38] S. Hou, X. Pan, C. C. Loy, Z. Wang, and D. Lin, "Learning a unified classifier incrementally via rebalancing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 831–839.
[39] S. Gidaris and N. Komodakis, "Dynamic few-shot visual learning without forgetting," 2018, *arXiv:1804.09458*.
[40] F. Cermelli, M. Mancini, S. R. Bulo, E. Ricci, and B. Caputo, "Modeling the background for incremental learning in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9230–9239.
[41] Y. Tian, Y. Wang, D. Krishnan, J. B. Tenenbaum, and P. Isola, "Rethinking few-shot image classification: A good embedding is all you need?" in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2020, pp. 266–282.
[42] I. Nigam, C. Huang, and D. Ramanan, "Ensemble knowledge transfer for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1499–1508.
[43] Z. Shao, K. Yang, and W. Zhou, "Performance evaluation of single-label and multi-label remote sensing image retrieval using a dense labeling dataset," *Remote Sens.*, vol. 10, no. 6, p. 964, Jun. 2018.
[44] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.
[45] *ISPRS 2D Semantic Labeling Contest*. Accessed: May 19, 2026. [Online]. Available: https://www.isprs.org/resources/datasets/benchmarks/Default.aspx
[46] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[47] H. Qi, M. Brown, and D. G. Lowe, "Low-shot learning with imprinted weights," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5822–5830.
[48] S. Gidaris and N. Komodakis, "Dynamic few-shot visual learning without forgetting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4367–4375.
[49] U. Michieli and P. Zanuttigh, "Incremental learning techniques for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3205–3212.
[50] A. Douillard, Y. Chen, A. Dapogny, and M. Cord, "PLOP: Learning without forgetting for continual semantic segmentation," 2020, *arXiv:2011.11390*.
[51] G. Park, W. Moon, S. Lee, T.-Y. Kim, and J.-P. Heo, "Mitigating background shift in class-incremental semantic segmentation," 2024, *arXiv:2407.11859*.
[52] Z. Xie, H. Lu, J. Xiao, E. Wang, L. Zhang, and X. Liu, "Early preparation pays off: New classifier pre-tuning for class incremental semantic segmentation," 2024, *arXiv:2407.14142*.
[53] M. Siam, B. Oreshkin, and M. Jagersand, "AMP: Adaptive masked proxies for few-shot segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5248–5257.
[54] F. Cermelli, M. Mancini, Y. Xian, Z. Akata, and B. Caputo, "Prototype-based incremental few-shot semantic segmentation," 2020, *arXiv:2012.01415*.
[55] R.-Z. Qiu, P. Chen, W. Sun, Y.-X. Wang, and K. Hauser, "GAPS: Few-shot incremental semantic segmentation via guided copy-paste synthesis," presented at the IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshop (CVPRW), Vancouver, BC, Canada, Jun. 2023. [Online]. Available: https://motion.cs.illinois.edu/papers/CVPRWorkshop2023-Qiu-FewShotSegmentation.pdf
[56] Y. Zhang, M. Shi, T. Su, and H. Wang, "Memory-based contrastive learning with optimized sampling for incremental few-shot semantic segmentation," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2024, pp. 1–5.



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